Electromagnetism — Problem Sheet 1

Worked solutions to Problems 1–20

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Questions and corrections are welcome by email.

How to use this document

Each problem starts with the question exactly as it is set on the sheet (orange box), including the sheet's own figure where there is one. The sheet marks problems as moderate difficulty (type 'A') or moderately demanding (type 'B'). Then follow:

- a **supervisor's note** (blue) on where the learning is and what students find conceptually hard;
- a **"What you need to know"** box (green) listing the background knowledge the solution relies on, with a pointer to where it is derived if it is not derived here;
- the **final answer** (grey), so that students can check their own work before reading on;
- the **worked solution**, with every algebraic step written out, including sign arguments, expansions, substitutions in integrals and changes between polar and Cartesian coordinates;
- **extensions** for students who want to go further (worked solutions are in a separate document).

SI units throughout. $\varepsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$ is the permittivity of free space, $1/(4\pi\varepsilon_0) = 8.988 \times 10^{9}\,\mathrm{m\,F^{-1}}$, and $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$ to the accuracy needed here. Every numerical answer has been checked by independent computation.

Toolkit used throughout the sheet

Binomial expansion. For $|u| \ll 1$, $(1+u)^n \approx 1 + nu + \frac{1}{2}n(n-1)u^2$. With $n = -\frac{1}{2}$: $(1+u)^{-1/2} \approx 1 - \frac{1}{2}u + \frac{3}{8}u^2$; with $n = -\frac{3}{2}$: $(1+u)^{-3/2} \approx 1 - \frac{3}{2}u$.
Spherical polar coordinates (r, θ, ϕ): θ is measured from the $+z$ axis and ϕ round it from the $+x$ axis.

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta, \qquad r^2 = x^2 + y^2 + z^2.$$

The unit vectors, obtained by differentiating the position vector $\mathbf{r} = r\,\hat{\mathbf{r}}$ with respect to each coordinate and normalising, are

$$\begin{aligned}
\hat{\mathbf{r}} &= \sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}},\\
\hat{\boldsymbol{\theta}} &= \frac{\partial\hat{\mathbf{r}}}{\partial\theta} = \cos\theta\cos\phi\,\hat{\mathbf{x}} + \cos\theta\sin\phi\,\hat{\mathbf{y}} - \sin\theta\,\hat{\mathbf{z}},\\
\hat{\boldsymbol{\phi}} &= -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}.
\end{aligned}$$

Note the *minus* sign in the $\hat{\mathbf{z}}$ part of $\hat{\boldsymbol{\theta}}$: increasing θ moves you *away* from the $+z$ axis, i.e. downwards. The gradient is $\nabla V = \dfrac{\partial V}{\partial r}\hat{\mathbf{r}} + \dfrac{1}{r}\dfrac{\partial V}{\partial\theta}\hat{\boldsymbol{\theta}} + \dfrac{1}{r\sin\theta}\dfrac{\partial V}{\partial\phi}\hat{\boldsymbol{\phi}}$.
Cylindrical polar coordinates (s, ϕ, z): $x = s\cos\phi$, $y = s\sin\phi$; $\hat{\mathbf{s}} = \cos\phi\,\hat{\mathbf{x}} + \sin\phi\,\hat{\mathbf{y}}$, $\hat{\boldsymbol{\phi}} = -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}$.

Electrostatics

Problem 1 — Finite line of charge

Problem 1 (A) (as set on the sheet)

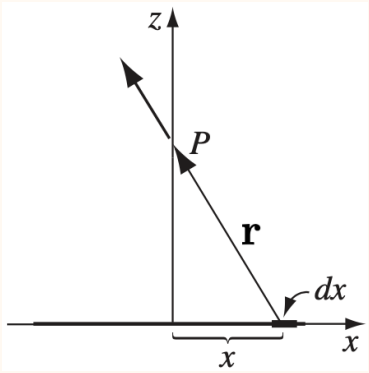
A line segment of length $2L$ carries a uniform charge per unit length λ.

(a) Find the electric field a distance z above the midpoint of the line segment, as shown in the diagram. *Hint: Use the symmetry of the system to argue that the* $\mathbf{E}$ *field is oriented along* $\hat{\mathbf{z}}$*, write down the contribution to* $\mathbf{E}$ *from the segment* $\mathrm{d}x$ *as shown, and integrate along* x*.*

(b) Consider a point far away from the line, i.e., in the limit $z \gg L$. Find the simplified expression for the electric field in this limit. Compare the result to a point charge and comment on your findings.

(c) Starting from the expression in 1a, find the electric field in the limit of $L \to \infty$.

(d) Find $\mathbf{E}$ in the limit of $L \to \infty$ using Gauss' law. Compare your results to 1c. Which of the two approaches do you find more elegant?



Supervisor's note

Where the learning is. Turning a symmetry argument into a calculation: identifying which components cancel *before* integrating, setting up $\mathrm{d}\mathbf{E}$ from a charge element, and reading limits physically. Part (d) is the first explicit contrast between brute-force integration and Gauss's law, and the moment to establish *when* Gauss's law can be used to calculate a field.

Conceptually tricky. Students often write the far-field correction as order L/z; recognising that the vanishing dipole moment makes it L^2/z^2 is the real insight in (b). In (d), many will invoke Gauss's law without justifying that $\mathbf{E}$ is radial and depends only on s — press them on what symmetry is being assumed, and why it fails for finite L.

What you need to know before starting

- **Coulomb's law for a small charge** $\mathrm{d}q$**:** $\mathrm{d}\mathbf{E} = \dfrac{\mathrm{d}q}{4\pi\varepsilon_0}\dfrac{\mathbf{r}}{|\mathbf{r}|^3}$, where $\mathbf{r}$ is the vector *from* the charge *to* the field point. (Writing $\mathbf{r}/|\mathbf{r}|^3 = \hat{\mathbf{r}}/r^2$ shows this is the familiar inverse-square law.) Fields from different elements add as vectors.
- **A trigonometric substitution** for integrals containing $(x^2+z^2)^{3/2}$, and the binomial expansion (toolkit, page 1).
- **Gauss's law** in integral form: $\oint_S \mathbf{E}\cdot\mathrm{d}\mathbf{A} = Q_{\mathrm{enc}}/\varepsilon_0$, where $\mathrm{d}\mathbf{A}$ points out of the closed surface S.

Final answer

$\mathbf{E} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2\lambda L}{z\sqrt{z^2+L^2}}\,\hat{\mathbf{z}}$; this tends to $Q/(4\pi\varepsilon_0 z^2)$ (with $Q = 2\lambda L$) far away and to $\lambda/(2\pi\varepsilon_0 z)$ for an infinite line.

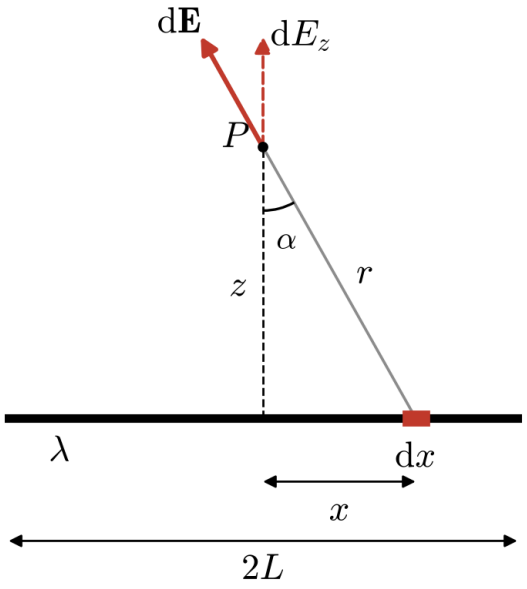


Figure 1: Geometry: the element dx at distance x from the centre of the segment, the field point P at height z on the bisector, and α, the angle between r and the perpendicular.

Set-up. Place the segment on the x-axis from $x = -L$ to $x = +L$, with the field point P at $(0, 0, z)$. The total charge is $Q = 2\lambda L$.

(a) Field on the perpendicular bisector

Step 1: the field of one element. The element between x and $x + \mathrm{d}x$ carries charge $\mathrm{d}q = \lambda\,\mathrm{d}x$ and sits at $(x, 0, 0)$. The vector from it to P is

$$\mathbf{r} = (0,0,z) - (x,0,0) = (-x,\,0,\,z), \qquad |\mathbf{r}| = \sqrt{x^2+z^2}.$$

Coulomb's law then gives the two components of its field:

$$\mathrm{d}E_x = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\,\mathrm{d}x\,(-x)}{(x^2+z^2)^{3/2}}, \qquad \mathrm{d}E_z = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\,\mathrm{d}x\,z}{(x^2+z^2)^{3/2}}.$$

(Equivalently, the field has magnitude $\lambda\,\mathrm{d}x/[4\pi\varepsilon_0(x^2+z^2)]$ and the fraction of it along $\hat{\mathbf{z}}$ is $\cos\alpha = z/\sqrt{x^2+z^2}$.)

Step 2: the symmetry argument, made explicit. The integrand of E_x is an *odd* function of x (it changes sign when $x \to -x$), and the range $-L \le x \le L$ is symmetric, so

$$E_x = -\frac{\lambda}{4\pi\varepsilon_0}\int_{-L}^{L}\frac{x\,\mathrm{d}x}{(x^2+z^2)^{3/2}} = 0.$$

Physically: the element at $+x$ pushes P up and to the left, its partner at $-x$ pushes it up and to the right, and the sideways pushes cancel. Only E_z remains.

Step 3: the integral for E_z.

$$E_z = \frac{\lambda z}{4\pi\varepsilon_0}\int_{-L}^{L}\frac{\mathrm{d}x}{(x^2+z^2)^{3/2}}.$$

Substitute $x = z\tan\alpha$ (α is the angle in the figure). Then

$$\mathrm{d}x = z\sec^2\alpha\,\mathrm{d}\alpha, \qquad x^2+z^2 = z^2(\tan^2\alpha+1) = z^2\sec^2\alpha, \qquad (x^2+z^2)^{3/2} = z^3\sec^3\alpha,$$

so

$$\int\frac{\mathrm{d}x}{(x^2+z^2)^{3/2}} = \int\frac{z\sec^2\alpha\,\mathrm{d}\alpha}{z^3\sec^3\alpha} = \frac{1}{z^2}\int\cos\alpha\,\mathrm{d}\alpha = \frac{\sin\alpha}{z^2}.$$

To return to x, draw the right-angled triangle with opposite side x and adjacent side z (so $\tan\alpha = x/z$); its hypotenuse is $\sqrt{x^2+z^2}$, so $\sin\alpha = x/\sqrt{x^2+z^2}$. Hence

$$\int_{-L}^{L}\frac{\mathrm{d}x}{(x^2+z^2)^{3/2}} = \left[\frac{x}{z^2\sqrt{x^2+z^2}}\right]_{-L}^{L} = \frac{L}{z^2\sqrt{L^2+z^2}} - \frac{-L}{z^2\sqrt{L^2+z^2}} = \frac{2L}{z^2\sqrt{L^2+z^2}}.$$

Step 4: assemble.

$$E_z = \frac{\lambda z}{4\pi\varepsilon_0}\cdot\frac{2L}{z^2\sqrt{z^2+L^2}} \quad\Longrightarrow\quad \boxed{\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{2\lambda L}{z\sqrt{z^2+L^2}}\,\hat{\mathbf{z}}}$$

For $\lambda > 0$ the field points away from the line (up, for $z > 0$). Dimensional check: λL is a charge and $z\sqrt{z^2+L^2}$ an area, so E has the form charge$/(4\pi\varepsilon_0 \times \text{length}^2)$, as it should.

(b) Far field, $z \gg L$

Take a factor z out of the square root so that the small quantity $u = L^2/z^2$ appears:

$$\sqrt{z^2+L^2} = z\sqrt{1+\frac{L^2}{z^2}} \quad\Longrightarrow\quad \frac{1}{z\sqrt{z^2+L^2}} = \frac{1}{z^2}\left(1+\frac{L^2}{z^2}\right)^{-1/2}.$$

The binomial expansion with $n = -\frac{1}{2}$ gives $(1+u)^{-1/2} \approx 1 - \frac{1}{2}u$, so

$$E \approx \frac{2\lambda L}{4\pi\varepsilon_0 z^2}\left(1-\frac{L^2}{2z^2}\right) = \frac{Q}{4\pi\varepsilon_0 z^2}\left(1-\frac{L^2}{2z^2}\right).$$

Comment. To leading order the segment looks like a point charge Q: from far away its extent is invisible. The first correction is of relative order L^2/z^2, not L/z. This is because the charge is distributed symmetrically about the origin, so it has no dipole moment; the leading departure from the point-charge field is a quadrupole term. The correction is negative because charge near the ends is farther from P than the centre is, and its field is tilted away from $\hat{\mathbf{z}}$.

(c) Infinite line, $L \to \infty$

Divide numerator and denominator by L:

$$\frac{2\lambda L}{z\sqrt{z^2+L^2}} = \frac{2\lambda}{z\sqrt{z^2/L^2+1}} \xrightarrow{L\to\infty} \frac{2\lambda}{z} \quad\Longrightarrow\quad E = \frac{\lambda}{2\pi\varepsilon_0 z}.$$

The field falls only as $1/z$. Moving away from the line, each element's field falls as $1/r^2$, but more and more of the line lies within a given angle of P, which partly compensates.

(d) Infinite line by Gauss's law

Symmetry first. An infinite uniform line looks the same after (i) sliding along itself and (ii) rotating about itself, and also after (iii) reflection in any plane perpendicular to it. Therefore $\mathbf{E}$ cannot depend on position along the line or angle round it, and it can have no component along the line (by (iii)) or round it (a circulating component would reverse under reflection in a plane containing the line). So $\mathbf{E} = E(s)\,\hat{\mathbf{s}}$, radial, where s is the perpendicular distance from the line.

Gaussian surface. Choose a closed cylinder of radius s and length ℓ coaxial with the line.

- Curved surface: $\mathrm{d}\mathbf{A}$ is along $\hat{\mathbf{s}}$, parallel to $\mathbf{E}$, and E is the same everywhere on it. Flux $= E(s) \times 2\pi s\ell$.
- Two flat end caps: $\mathrm{d}\mathbf{A}$ is along the line, perpendicular to $\mathbf{E}$. Flux $= 0$.

The enclosed charge is $\lambda\ell$, so

$$E(s)\,2\pi s\ell = \frac{\lambda\ell}{\varepsilon_0} \quad\Longrightarrow\quad E = \frac{\lambda}{2\pi\varepsilon_0 s},$$

identical to (c) with $s = z$.

Which is more elegant? Gauss's law takes one line, but only because the symmetry is perfect. It cannot give (a) or (b): for a finite segment there is no surface on which E is both constant and normal, so the flux integral cannot be turned into "$E\times$area". (Gauss's law is still *true* for the finite segment; it just doesn't determine $\mathbf{E}$.) Direct integration is longer but works for any geometry.

Extensions

1. Find $\mathbf{E}$ at a general point off the bisector; express it in terms of the angles subtended by the two ends of the segment.
2. Show that the field of a uniformly charged ring on its axis is obtained by the same method, and compare its far-field correction with that of the segment.
3. Sketch the field lines near the end of a finite segment: where does the infinite-line approximation fail?

Problem 2 — Charged sphere, the Earth's charge, and a finite capacitor

Problem 2 (A) (as set on the sheet)

(a) A conducting sphere of radius 10 cm is charged to 5 kV above earth potential and isolated. Find the charge on the sphere, the E-field just outside it, and the field and potential 20 cm from its centre.

(b) Estimate the total charge on the Earth if the mean atmospheric E-field near the surface is $100\,\mathrm{V\,m^{-1}}$. [The radius of the Earth is about 6400 km.]

(c) Draw a simple sketch of the electric field lines and equipotentials around an isolated parallel-plate capacitor with equal and opposite charge on the two plates. Take the plates to be infinite strips about 5 times wider than their separation.

Supervisor's note

Where the learning is. Capacitance of an isolated conductor, and fluency with orders of magnitude (fields, charges, breakdown). Part (c) tests whether students can construct a

field picture from first principles: field lines orthogonal to equipotentials, fringing, and the structure of the external field.

Conceptually tricky. In (b) almost everyone gives a positive number; the sign follows from the direction of the fair-weather field. In (c), sketches typically show only the uniform interior field and miss the fringing at the edges, the weak external field looping from the outer face of $+$ to $-$, and the dipole-like far field.

What you need to know before starting

- **Field of a spherically symmetric charge.** Gauss's law on a sphere of radius r gives $E \cdot 4\pi r^2 = Q/\varepsilon_0$ outside the charge, so the field outside is that of a point charge Q at the centre.
- **Potential from field:** $V(r) = \int_r^\infty E(r')\,\mathrm{d}r'$, taking $V(\infty) = 0$ ("earth potential" is taken as the zero, i.e. the same as at infinity).
- **Capacitance** $C = Q/V$. Field lines start on $+$ charge and end on $-$ charge; they cross equipotentials at right angles.
- The facts used about conductors (charge on the surface, **E** normal to it) are proved in Problem 6.

Final answer

(a) $Q = 5.56 \times 10^{-8}\,\mathrm{C}$; $E = 50\,\mathrm{kV\,m^{-1}}$ at the surface; at 20 cm, $V = 2.5\,\mathrm{kV}$ and $E = 12.5\,\mathrm{kV\,m^{-1}}$. (b) $Q_\oplus \approx -4.6 \times 10^5\,\mathrm{C}$. (c) Figure 2.

(a) Conducting sphere at $5\,\mathrm{kV}$

Step 1: field and potential outside. The charge Q sits uniformly on the surface of the sphere (Problem 6). By Gauss's law, for $r \geq R$,

$$E(r) = \frac{Q}{4\pi\varepsilon_0 r^2}, \qquad V(r) = \int_r^\infty \frac{Q}{4\pi\varepsilon_0 r'^2}\,\mathrm{d}r' = \left[-\frac{Q}{4\pi\varepsilon_0 r'}\right]_r^\infty = \frac{Q}{4\pi\varepsilon_0 r}.$$

Step 2: the charge. At the surface $r = R = 0.10\,\mathrm{m}$ the potential is $V_0 = 5.0\,\mathrm{kV}$:

$$V_0 = \frac{Q}{4\pi\varepsilon_0 R} \quad\Longrightarrow\quad Q = 4\pi\varepsilon_0 R\, V_0.$$

(This says that an isolated sphere has capacitance $C = Q/V_0 = 4\pi\varepsilon_0 R$.) Numerically, with $4\pi\varepsilon_0 = 1.1127 \times 10^{-10}\,\mathrm{F\,m^{-1}}$,

$$Q = (1.1127 \times 10^{-10}\,\mathrm{F\,m^{-1}})(0.10\,\mathrm{m})(5.0 \times 10^3\,\mathrm{V}) = 5.56 \times 10^{-8}\,\mathrm{C}.$$

Step 3: the surface field. Dividing $E(R)$ by $V(R)$ gives $E(R)/V_0 = 1/R$, so

$$E(R) = \frac{V_0}{R} = \frac{5.0 \times 10^3\,\mathrm{V}}{0.10\,\mathrm{m}} = 5.0 \times 10^4\,\mathrm{V\,m^{-1}} = 50\,\mathrm{kV\,m^{-1}}.$$

Step 4: at $r = 20\,\mathrm{cm} = 2R$**.** Since $V \propto 1/r$ and $E \propto 1/r^2$,

$$V(2R) = \frac{V_0}{2} = 2.5\,\mathrm{kV}, \qquad E(2R) = \frac{E(R)}{4} = 12.5\,\mathrm{kV\,m^{-1}}.$$

Checks. The printed $5.5606 \times 10^{-8}\,\mathrm{C}$ uses $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$; the CODATA value gives $5.5633 \times 10^{-8}\,\mathrm{C}$, so quote $5.56 \times 10^{-8}\,\mathrm{C}$. And $50\,\mathrm{kV\,m^{-1}}$ is far below the breakdown field of dry air ($\approx 3\,\mathrm{MV\,m^{-1}}$), so the sphere can indeed hold this charge.

(b) Charge on the Earth

Treat the Earth as a conducting sphere of radius $R_\oplus = 6.4 \times 10^6\,\mathrm{m}$. Just outside it, $E = |Q|/(4\pi\varepsilon_0 R_\oplus^2)$, so

$$\begin{aligned} |Q| = 4\pi\varepsilon_0 R_\oplus^2 E &= (1.1127 \times 10^{-10}\,\mathrm{F\,m^{-1}})(6.4 \times 10^6\,\mathrm{m})^2(100\,\mathrm{V\,m^{-1}}) \\ &= (1.1127 \times 10^{-10})(4.096 \times 10^{13})(100)\ \mathrm{C} = 4.6 \times 10^5\,\mathrm{C}. \end{aligned}$$

The sign. The fair-weather field points *downwards*, towards the ground. Field lines end on negative charge, so the Earth's charge is negative: $Q_\oplus \approx -4.6 \times 10^5\,\mathrm{C}$. The surface density is $\sigma = \varepsilon_0 E_\perp \approx -(8.85 \times 10^{-12})(100) = -8.9 \times 10^{-10}\,\mathrm{C\,m^{-2}}$, roughly one electron per $180\,\mu\mathrm{m}^2$.

This is an estimate: the compensating positive charge lies in the atmosphere (the "global circuit"), and the field varies strongly with weather; thunderstorms keep the Earth charged (compare Problem 8).

(c) Field lines and equipotentials of a finite parallel-plate capacitor

Figure 2 is a numerical solution of Laplace's equation $\nabla^2 V = 0$ in two dimensions for two strips of width $w = 5d$ at potentials $\pm V/2$, carrying $+Q$ (top) and $-Q$ (bottom); d is the separation. A hand sketch only needs the features listed below.

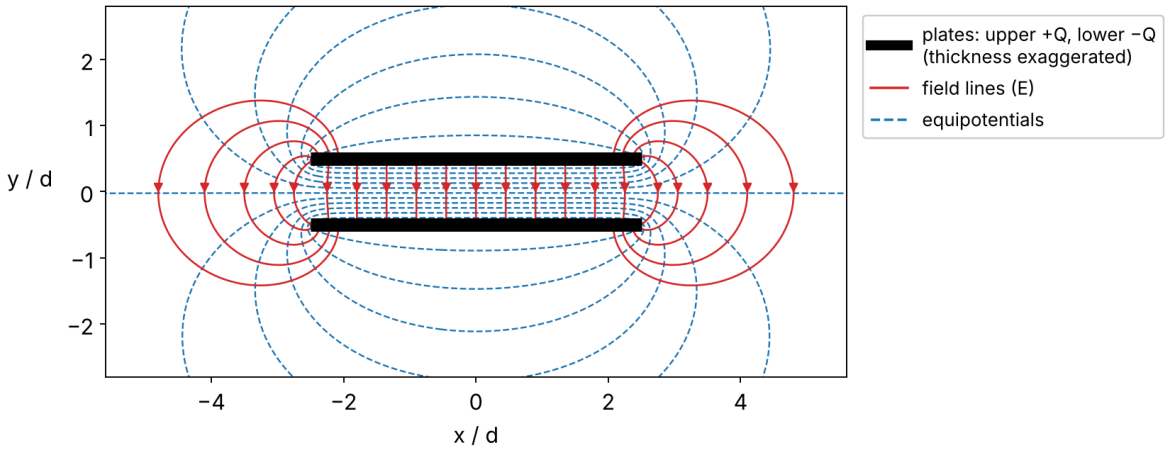


Figure 2: Computed field lines (solid) and equipotentials (dashed) around an isolated parallel-plate capacitor whose plates are five times wider than their separation d.

A good sketch should show:

- **Interior:** straight, parallel, evenly spaced field lines from $+$ to $-$, and evenly spaced plane equipotentials: the uniform field $E = \sigma/\varepsilon_0 = V/d$.
- **Edges (fringing):** field lines bulge outwards and equipotentials wrap round the plate ends; the field is strongest at the sharp edges, where charge accumulates.
- **Outside:** weak field; lines leave the outer face of the $+$ plate near its edges and loop round to the outer face of the $-$ plate. Almost all the charge sits on the inner faces.
- **Symmetry:** the mid-plane is the $V = 0$ equipotential and every field line crosses it at right angles.
- **Far away:** equipotentials close around the capacitor. The net charge is zero, so the far field is that of a (line) dipole.

- **Everywhere:** field lines cut equipotentials at right angles and begin and end only on charges.

Extensions

1. What radius of sphere would be needed to hold 5 kV before air breaks down at 3 MV m^{-1}? Relate this to why high-voltage terminals are large and smooth.
2. Use the Earth's charge from (b) and a fair-weather current density of $\sim 2 \times 10^{-12}$ A m^{-2} to estimate how long the Earth would take to discharge without thunderstorms.
3. In (c), estimate the fractional correction to the ideal capacitance $\varepsilon_0 A/d$ caused by fringing.

Problem 3 — Field of a small dipole in Cartesian components

Problem 3 (A) (as set on the sheet)

A small dipole centred at the origin of coordinates has Cartesian vector components $(0, 0, p)$, where $p = qa$, q is the charge, and a the separation. By differentiating the expression for the potential at point (x, y, z), find the x, y and z components of the electric field when $(x^2 + y^2 + z^2) \gg a^2$.

Supervisor's note

Where the learning is. Deriving the dipole potential by a controlled expansion in a/r, then using $\mathbf{E} = -\nabla V$ in Cartesian coordinates. This is the first real use of the multipole idea and of the gradient as a computational tool rather than a definition.

Conceptually tricky. Students lose E_z by forgetting that z appears both explicitly and inside r (product rule). Ask them to check the axial and equatorial limits and to recover the coordinate-free form $[3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]/r^3$ — this is the form they will need in Problems 4, 7 and 20.

What you need to know before starting

- **Potential of a point charge**, $V = q/(4\pi\varepsilon_0 r)$, and superposition of potentials (they add as scalars).
- $\mathbf{E} = -\nabla V = -\left(\dfrac{\partial V}{\partial x}, \dfrac{\partial V}{\partial y}, \dfrac{\partial V}{\partial z}\right)$.
- **Partial derivatives with the chain and product rules**, the binomial expansion, and spherical polar coordinates (all in the toolkit on page 1).

Final answer

With $r^2 = x^2 + y^2 + z^2$: $\quad (E_x, E_y, E_z) = \dfrac{p}{4\pi\varepsilon_0 r^5}\left(3xz,\ 3yz,\ 3z^2 - r^2\right)$.

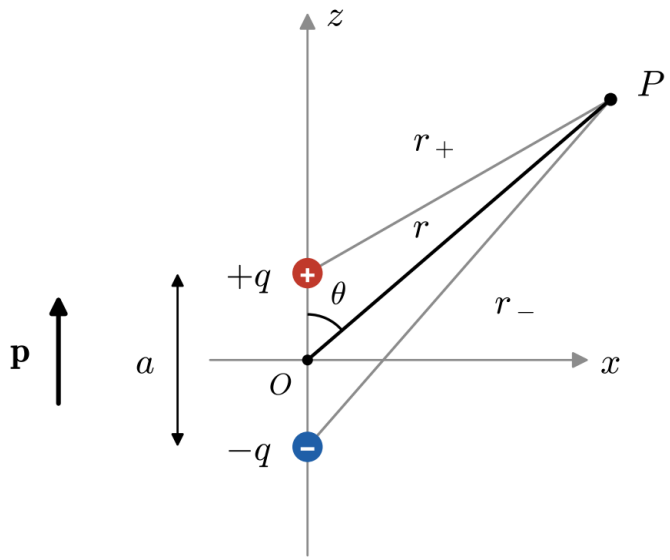


Figure 3: Dipole geometry: charges $\pm q$ a distance a apart along z; the field point P is at distance r from the centre and polar angle θ from the dipole axis.

Set-up. Put $+q$ at $(0,0,a/2)$ and $-q$ at $(0,0,-a/2)$, so $\mathbf{p} = qa\,\hat{\mathbf{z}}$ (the dipole moment points from $-q$ to $+q$).

Step 1: the exact potential

The distances from the two charges to $P = (x,y,z)$ are r_+ and r_-, with

$$r_\pm^2 = x^2 + y^2 + \left(z \mp \tfrac{a}{2}\right)^2 = \underbrace{x^2+y^2+z^2}_{r^2} \mp az + \tfrac{a^2}{4}.$$

Adding the two point-charge potentials,

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_+} - \frac{1}{r_-}\right).$$

Step 2: expand for $r \gg a$

Factor out r^2:

$$r_\pm^2 = r^2\left(1 \mp \frac{az}{r^2} + \frac{a^2}{4r^2}\right) \quad\Longrightarrow\quad \frac{1}{r_\pm} = \frac{1}{r}\left(1 \mp \frac{az}{r^2} + \frac{a^2}{4r^2}\right)^{-1/2}.$$

Since $|z| \le r$, the term az/r^2 is at most of order a/r, while $a^2/(4r^2)$ is of order $(a/r)^2$. Keeping only first order in a/r and using $(1+u)^{-1/2} \approx 1 - \tfrac{1}{2}u$ with $u = \mp az/r^2$:

$$\frac{1}{r_\pm} \approx \frac{1}{r}\left(1 \pm \frac{az}{2r^2}\right).$$

Subtract:

$$\frac{1}{r_+} - \frac{1}{r_-} \approx \frac{1}{r}\left[\left(1 + \frac{az}{2r^2}\right) - \left(1 - \frac{az}{2r^2}\right)\right] = \frac{az}{r^3}.$$

Therefore

$$\boxed{V = \frac{qa\,z}{4\pi\varepsilon_0 r^3} = \frac{p\,z}{4\pi\varepsilon_0 r^3}} \qquad \text{or, using } z = r\cos\theta, \qquad V = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}.$$

The terms of order $(a/r)^2$ that we dropped are the same for r_+ and r_- and cancel in the difference, so the first neglected term in V is smaller than the one kept by a factor of order $(a/r)^2$.

Step 3: differentiate, $\mathbf{E} = -\nabla V$

Write $K = p/(4\pi\varepsilon_0)$, so $V = K\,z\,r^{-3}$, and note that r depends on x, y and z. First,

$$r = (x^2+y^2+z^2)^{1/2} \quad\Longrightarrow\quad \frac{\partial r}{\partial x} = \tfrac{1}{2}(x^2+y^2+z^2)^{-1/2}\cdot 2x = \frac{x}{r},$$

and similarly $\partial r/\partial y = y/r$, $\partial r/\partial z = z/r$. By the chain rule,

$$\frac{\partial (r^{-3})}{\partial x} = -3r^{-4}\,\frac{\partial r}{\partial x} = -3r^{-4}\,\frac{x}{r} = -\frac{3x}{r^5}, \qquad \text{and likewise} \quad \frac{\partial (r^{-3})}{\partial y} = -\frac{3y}{r^5}, \quad \frac{\partial (r^{-3})}{\partial z} = -\frac{3z}{r^5}.$$

x- and y-components. Here z is a constant factor:

$$E_x = -\frac{\partial V}{\partial x} = -Kz\left(-\frac{3x}{r^5}\right) = \frac{3p\,xz}{4\pi\varepsilon_0 r^5}, \qquad E_y = -Kz\left(-\frac{3y}{r^5}\right) = \frac{3p\,yz}{4\pi\varepsilon_0 r^5}.$$

z-component. Now z appears twice, explicitly and inside r, so use the product rule:

$$\frac{\partial}{\partial z}\big(z\,r^{-3}\big) = 1\cdot r^{-3} + z\cdot\left(-\frac{3z}{r^5}\right) = \frac{r^2-3z^2}{r^5}, \qquad E_z = -K\,\frac{r^2-3z^2}{r^5} = \frac{p\,(3z^2-r^2)}{4\pi\varepsilon_0 r^5}.$$

Step 4: the same field in spherical polars, and back to Cartesians

It is worth seeing the same result in polar form, which is what Problems 4, 7 and 20 use. With $V = p\cos\theta/(4\pi\varepsilon_0 r^2)$ and the polar gradient (toolkit):

$$E_r = -\frac{\partial V}{\partial r} = \frac{2p\cos\theta}{4\pi\varepsilon_0 r^3}, \qquad E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta} = \frac{p\sin\theta}{4\pi\varepsilon_0 r^3}, \qquad E_\phi = 0.$$

Converting to Cartesians. $\mathbf{E} = E_r\hat{\mathbf{r}} + E_\theta\hat{\boldsymbol{\theta}}$. Substitute the Cartesian forms of $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ and collect the coefficient of each Cartesian unit vector:

$$E_z = E_r\cos\theta + E_\theta(-\sin\theta) = \frac{p}{4\pi\varepsilon_0 r^3}\big(2\cos^2\theta - \sin^2\theta\big) = \frac{p}{4\pi\varepsilon_0 r^3}\big(3\cos^2\theta - 1\big),$$

$$E_x = E_r\sin\theta\cos\phi + E_\theta\cos\theta\cos\phi = \frac{p}{4\pi\varepsilon_0 r^3}\big(2+1\big)\sin\theta\cos\theta\cos\phi = \frac{3p\sin\theta\cos\theta\cos\phi}{4\pi\varepsilon_0 r^3},$$

and E_y is the same with $\cos\phi \to \sin\phi$. Now use $\cos\theta = z/r$ and $\sin\theta\cos\phi = x/r$, $\sin\theta\sin\phi = y/r$:

$$E_z = \frac{p}{4\pi\varepsilon_0 r^3}\left(\frac{3z^2}{r^2} - 1\right) = \frac{p(3z^2-r^2)}{4\pi\varepsilon_0 r^5}, \qquad E_x = \frac{3p}{4\pi\varepsilon_0 r^3}\cdot\frac{x}{r}\cdot\frac{z}{r} = \frac{3p\,xz}{4\pi\varepsilon_0 r^5},$$

in agreement with Step 3.

Checks

- **On the axis** ($x = y = 0$, so $r = |z|$): $E_x = E_y = 0$ and $E_z = p(3z^2-z^2)/(4\pi\varepsilon_0|z|^5) = 2p/(4\pi\varepsilon_0|z|^3)$, along $+\hat{\mathbf{z}}$, i.e. *parallel* to $\mathbf{p}$, on both sides of the dipole.
- **In the equatorial plane** ($z = 0$): $E_x = E_y = 0$ and $E_z = -p\,r^2/(4\pi\varepsilon_0 r^5) = -p/(4\pi\varepsilon_0 r^3)$: *antiparallel* to $\mathbf{p}$ and half the axial strength at the same distance. (In polar language: at $\theta = 90^\circ$, $E_r = 0$ and $E_\theta > 0$, and there $\hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}$.)
- $\nabla\cdot\mathbf{E} = 0$ for $r \neq 0$, as required in empty space.
- **Coordinate-free form**, often the most useful: since $\mathbf{p}\cdot\hat{\mathbf{r}} = p\cos\theta = pz/r$,

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\,\frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3}.$$

  Check the z-component: $[3(pz/r)(z/r) - p]/r^3 = p(3z^2-r^2)/r^5$.

Extensions

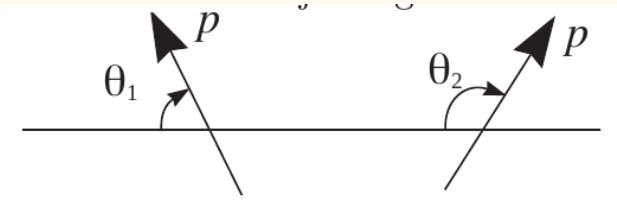
1. Compute the quadrupole potential of a linear arrangement $+q$, $-2q$, $+q$ and compare its fall-off with the dipole.
2. Find the equation of the dipole field lines in polar coordinates ($r \propto \sin^2\theta$) and sketch them.
3. Show directly that $\nabla \cdot \mathbf{E} = 0$ and $\nabla \times \mathbf{E} = 0$ for the field found here.

Problem 4 — Interaction energy of two coplanar dipoles

Problem 4 (B) (as set on the sheet)

Two coplanar identical small electric dipoles of moment p are supported on pivots a large distance d apart. Each dipole can rotate only in the plane. Their angles of twist, θ_1 and θ_2, are measured clockwise from the line joining their centres as shown in the diagram. Show, by treating each dipole as the sum of two components at right angles, that the potential energy of one in the field of the other is

$$-\frac{p^2}{8\pi\varepsilon_0 d^3}\big(3\cos(\theta_1+\theta_2)+\cos(\theta_1-\theta_2)\big).$$



Supervisor's note

Where the learning is. Superposing fields of dipole components, and the energy of a dipole in an external field, $U = -\mathbf{p}\cdot\mathbf{E}$. Decomposing into axial and transverse pieces is a transferable technique (it reappears for magnetic dipoles in Problem 20).

Conceptually tricky. Sign conventions are the trap: the equatorial field is *antiparallel* to the dipole, and the angles must be measured in the same sense for both dipoles. A single sign slip swaps $\theta_1+\theta_2$ and $\theta_1-\theta_2$. Ask students to classify the equilibria — most will not spot that the antiparallel side-by-side state is a saddle point, not a minimum.

What you need to know before starting

- **The dipole field** from Problem 3, especially its values on the axis (parallel to $\mathbf{p}$, magnitude $2p/4\pi\varepsilon_0 r^3$) and in the equatorial plane (antiparallel to $\mathbf{p}$, magnitude $p/4\pi\varepsilon_0 r^3$). Step 1 below shows where both come from.
- **The energy of a dipole in an external field**, $U = -\mathbf{p}\cdot\mathbf{E}$, derived in Step 3.
- **Compound-angle formulae**: $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$.

Final answer

$$U = -\frac{p^2}{8\pi\varepsilon_0 d^3}\big[3\cos(\theta_1+\theta_2)+\cos(\theta_1-\theta_2)\big].$$

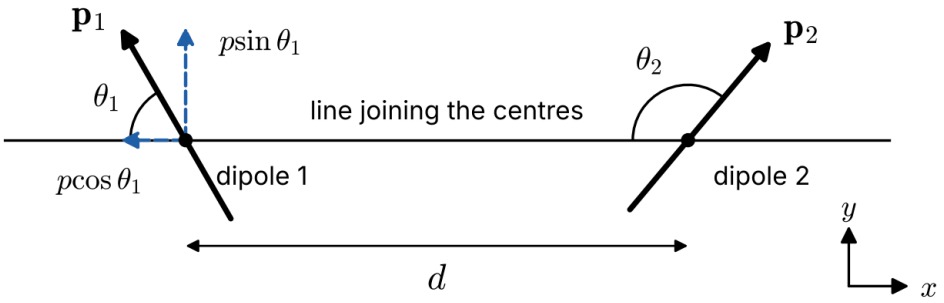


Figure 4: Both angles are measured in the same sense from the line joining the dipoles, a distance d apart. Dipole 1 is resolved into its axial part $p\cos\theta_1$ and transverse part $p\sin\theta_1$.

Step 0: axes, and the two components of each dipole

Use ordinary Cartesian axes in the plane of the diagram: $\hat{\mathbf{x}}$ along the line joining the centres, pointing *from dipole 1 to dipole 2* (to the right), and $\hat{\mathbf{y}}$ perpendicular to it, upwards. Dipole 1 is at the origin and dipole 2 at $(d, 0)$.

Reading the angles off the diagram. Each angle is measured from the line, starting from the *leftward* direction and turning clockwise, i.e. up and over. So:

- at $\theta = 0$ the dipole points left, along $-\hat{\mathbf{x}}$;
- at $\theta = 90^\circ$ it points straight up, along $+\hat{\mathbf{y}}$;
- at $\theta = 180^\circ$ it points right, along $+\hat{\mathbf{x}}$.

A dipole that satisfies all three is

$$\mathbf{p}_i = p_{ix}\,\hat{\mathbf{x}} + p_{iy}\,\hat{\mathbf{y}} \qquad \text{with} \qquad p_{ix} = -p\cos\theta_i, \quad p_{iy} = +p\sin\theta_i \qquad (i = 1, 2).$$

(Check against the diagram: θ_1 is acute, so $\mathbf{p}_1$ points up and to the left; θ_2 is obtuse, so $-\cos\theta_2 > 0$ and $\mathbf{p}_2$ points up and to the right. The arrow labelled $p\cos\theta_1$ in the figure is the component of length $p\cos\theta_1$ pointing left, i.e. $p_{1x} = -p\cos\theta_1$.)

These are the "two components at right angles" of the question: p_{ix} lies *along* the line joining the dipoles, and p_{iy} is *perpendicular* to it.

Step 1: two facts about the dipole field, from Problem 3

Problem 3 found the field of a dipole $p\,\hat{\mathbf{z}}$ at the origin:

$$(E_x, E_y, E_z) = \frac{p}{4\pi\varepsilon_0 r^5}\left(3xz,\ 3yz,\ 3z^2 - r^2\right).$$

We need only two special places.

(a) On the dipole's own axis ($x = y = 0$, so $r = |z|$):

$$E_x = E_y = 0, \qquad E_z = \frac{p\,(3z^2 - z^2)}{4\pi\varepsilon_0 |z|^5} = +\frac{2p}{4\pi\varepsilon_0 |z|^3}.$$

E_z is positive whether z is positive or negative: *on the axis, on either side, the field points the same way as the dipole*, with magnitude $2p/(4\pi\varepsilon_0 r^3)$.

(b) In the dipole's equatorial plane ($z = 0$, i.e. on any line through the dipole at right angles to it):

$$E_x = E_y = 0, \qquad E_z = \frac{p\,(0 - r^2)}{4\pi\varepsilon_0 r^5} = -\frac{p}{4\pi\varepsilon_0 r^3}.$$

This is where the minus sign comes from: with $z = 0$ the bracket $3z^2 - r^2$ is just $-r^2$. *At the side of a dipole the field points opposite to the dipole*, with half the axial magnitude.

Picture. At a point beside the dipole, $+q$ and $-q$ are equally far away. The field of $+q$ points away from $+q$; the field of $-q$ points towards $-q$. Their sideways parts cancel, and the parts parallel to the dipole both point from the $+$ end towards the $-$ end, i.e. *against* $\mathbf{p}$. (Field lines leave the $+$ end, loop round, and return to the $-$ end; beside the dipole they are travelling backwards.)

These two facts do not depend on how the axes are labelled: (a) says "along $\mathbf{p}$" and (b) says "against $\mathbf{p}$", whatever direction $\mathbf{p}$ points in.

Step 2: field of dipole 1 at dipole 2

Treat dipole 1 as two dipoles, $p_{1x}\hat{\mathbf{x}}$ and $p_{1y}\hat{\mathbf{y}}$, and add their fields at dipole 2, which is at distance d along $+\hat{\mathbf{x}}$.

- $p_{1x}\hat{\mathbf{x}}$ points along the line joining the dipoles, so dipole 2 is **on its axis**. By (a) its field there is along $p_{1x}\hat{\mathbf{x}}$ with magnitude $2/(4\pi\varepsilon_0 d^3)$ times its strength:

$$\mathbf{E}_{(x\text{-part})} = \frac{2p_{1x}}{4\pi\varepsilon_0 d^3}\,\hat{\mathbf{x}}.$$

- $p_{1y}\hat{\mathbf{y}}$ is perpendicular to the line joining the dipoles, so dipole 2 is **in its equatorial plane**. By (b) its field there is *opposite* to $p_{1y}\hat{\mathbf{y}}$:

$$\mathbf{E}_{(y\text{-part})} = -\frac{p_{1y}}{4\pi\varepsilon_0 d^3}\,\hat{\mathbf{y}}.$$

(If p_{1x} or p_{1y} is negative, these formulas still hold: the field simply reverses with the component.) Adding,

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0 d^3}\big(2p_{1x}\,\hat{\mathbf{x}} - p_{1y}\,\hat{\mathbf{y}}\big).$$

Step 3: energy of dipole 2 in this field

Model dipole 2 as $+q$ at $\mathbf{r} + \mathbf{a}/2$ and $-q$ at $\mathbf{r} - \mathbf{a}/2$, with $\mathbf{p}_2 = q\mathbf{a}$. Its energy in an external potential V is

$$U = qV\big(\mathbf{r} + \tfrac{\mathbf{a}}{2}\big) - qV\big(\mathbf{r} - \tfrac{\mathbf{a}}{2}\big) \approx q\left[V + \tfrac{\mathbf{a}}{2}\cdot\nabla V\right] - q\left[V - \tfrac{\mathbf{a}}{2}\cdot\nabla V\right] = q\mathbf{a}\cdot\nabla V = -\mathbf{p}_2\cdot\mathbf{E},$$

using a first-order Taylor expansion of V (valid because $d \gg a$, so $\mathbf{E}$ is uniform across the dipole). The dot product is just "x times x plus y times y":

$$U = -\big(p_{2x}E_x + p_{2y}E_y\big) = -\frac{1}{4\pi\varepsilon_0 d^3}\big(2p_{1x}p_{2x} - p_{1y}p_{2y}\big).$$

Step 4: put in the angles

From Step 0,

$$p_{1x}p_{2x} = (-p\cos\theta_1)(-p\cos\theta_2) = p^2\cos\theta_1\cos\theta_2,$$
$$p_{1y}p_{2y} = (p\sin\theta_1)(p\sin\theta_2) = p^2\sin\theta_1\sin\theta_2.$$

(The two minus signs in the x-components cancel, so it does not matter that the angles are measured from the left.) Therefore

$$U = -\frac{p^2}{4\pi\varepsilon_0 d^3}\big(2\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2\big).$$

Step 5: convert to sum and difference angles

From $\cos(\theta_1 \pm \theta_2) = \cos\theta_1\cos\theta_2 \mp \sin\theta_1\sin\theta_2$, adding and subtracting the two versions gives

$$\cos\theta_1\cos\theta_2 = \tfrac{1}{2}\big[\cos(\theta_1-\theta_2)+\cos(\theta_1+\theta_2)\big], \qquad \sin\theta_1\sin\theta_2 = \tfrac{1}{2}\big[\cos(\theta_1-\theta_2)-\cos(\theta_1+\theta_2)\big].$$

Therefore

$$\begin{aligned} 2\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 &= \big[\cos(\theta_1-\theta_2)+\cos(\theta_1+\theta_2)\big] - \tfrac{1}{2}\big[\cos(\theta_1-\theta_2)-\cos(\theta_1+\theta_2)\big] \\ &= \tfrac{3}{2}\cos(\theta_1+\theta_2) + \tfrac{1}{2}\cos(\theta_1-\theta_2), \end{aligned}$$

and

$$\boxed{U = -\frac{p^2}{8\pi\varepsilon_0 d^3}\big[3\cos(\theta_1+\theta_2)+\cos(\theta_1-\theta_2)\big]}$$

The result is symmetric in θ_1 and θ_2, as a mutual energy must be: we would get the same answer by finding the field of dipole 2 at dipole 1.

Check with the general formula. The interaction energy of two dipoles separated by $\mathbf{d}$ is $U = [\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{p}_1\cdot\hat{\mathbf{d}})(\mathbf{p}_2\cdot\hat{\mathbf{d}})]/(4\pi\varepsilon_0 d^3)$. Here $\hat{\mathbf{d}} = \hat{\mathbf{x}}$, so $\mathbf{p}_1\cdot\mathbf{p}_2 = p_{1x}p_{2x} + p_{1y}p_{2y}$ and $(\mathbf{p}_1\cdot\hat{\mathbf{d}})(\mathbf{p}_2\cdot\hat{\mathbf{d}}) = p_{1x}p_{2x}$, giving $U = (p_{1y}p_{2y} - 2p_{1x}p_{2x})/(4\pi\varepsilon_0 d^3)$, the same as Step 3.

Physical checks

Configuration	θ_1, θ_2	$U \big/ \frac{p^2}{4\pi\varepsilon_0 d^3}$	Comment
Head-to-tail along the line	0, 0	-2	Lowest energy: stable ground state
Head-to-head along the line	0, π	$+2$	Highest energy
Side by side, parallel	$\pi/2$, $\pi/2$	$+1$	Like charges level: repulsive
Side by side, antiparallel	$\pi/2$, $-\pi/2$	-1	Saddle point (see below)

For example, for $(\pi/2, \pi/2)$: $3\cos\pi + \cos 0 = -2$, so $U = +2p^2/(8\pi\varepsilon_0 d^3) = +p^2/(4\pi\varepsilon_0 d^3)$. The antiparallel side-by-side state is stable against rotating both dipoles together but unstable against counter-rotation. Free dipoles therefore prefer to line up head-to-tail along the line joining them, which is why polar molecules form chains.

Extensions

1. Find and classify all equilibrium orientations when both dipoles are free to rotate (minima, maxima, saddle points).
2. Find the torque on dipole 2 as a function of θ_1 and θ_2, and show that the torques on the two dipoles are not equal and opposite. Where does the missing angular momentum go?
3. Estimate the interaction energy of two water molecules ($p \approx 6.2\times10^{-30}\,\mathrm{C\,m}$) at 0.3 nm and compare with $k_B T$ at room temperature.

Problem 5 — Self-energy of a uniform sphere; classical electron radius

Problem 5 (A) (as set on the sheet)

Calculate the electrostatic energy $U = \frac{1}{2}\int \mathrm{d}\tau\, \rho V$ of a charge Q uniformly distributed throughout a sphere of radius a. Hence calculate a "classical radius" of the electron on the dubious assumption that its rest mass is due to the electrostatic energy.

Supervisor's note

Where the learning is. Electrostatic energy computed three equivalent ways — $\frac{1}{2}\int \rho V\,\mathrm{d}\tau$, $\frac{\varepsilon_0}{2}\int E^2\,\mathrm{d}\tau$, and assembling shell by shell — and seeing that they agree. The "classical electron radius" is a vehicle for discussing where classical electromagnetism breaks down.

Conceptually tricky. The origin of the factor $\frac{1}{2}$ (double counting of pairs) is widely misunderstood. Students also tend to take the classical radius literally; the discussion of why the assumption is "dubious" (model-dependent prefactor, divergent self-energy of a point charge, stability) is where the physics is.

What you need to know before starting

- **Why $U = \frac{1}{2}\int \rho V\,\mathrm{d}\tau$.** For point charges, $U = \sum_{\text{pairs}} q_i q_j/(4\pi\varepsilon_0 r_{ij}) = \frac{1}{2}\sum_i q_i V_i$, where V_i is the potential at q_i due to all the others. The sum over i counts each pair twice, hence the $\frac{1}{2}$. The integral is the continuous version.
- **Gauss's law for a sphere**, and $V(r) = \int_r^\infty E\,\mathrm{d}r'$.
- **Volume element for spherical symmetry:** a thin shell of radius r and thickness $\mathrm{d}r$ has volume $\mathrm{d}\tau = 4\pi r^2\,\mathrm{d}r$.
- **Mass–energy equivalence**, $E = mc^2$ (special relativity).

Final answer

$U = \dfrac{3Q^2}{20\pi\varepsilon_0 a}$; equating U to $m_e c^2$ gives $a = 1.69\times10^{-15}\,\mathrm{m}$.

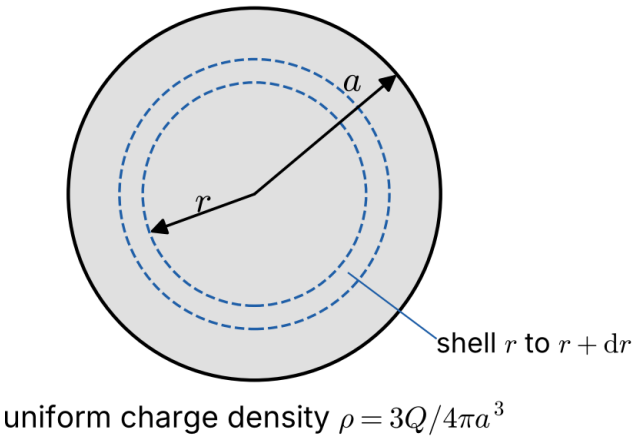


Figure 5: Uniformly charged sphere of radius a, with the thin shell between r and $r + \mathrm{d}r$ used in the energy integral.

Set-up. Charge Q fills a sphere of radius a uniformly, so the charge density is $\rho = Q/\frac{4}{3}\pi a^3 = 3Q/(4\pi a^3)$.

Step 1: the field (Gauss's law)

Take a Gaussian sphere of radius r. By symmetry $\mathbf{E}$ is radial with magnitude $E(r)$, so the flux is $E \cdot 4\pi r^2$.

- Inside ($r \le a$): the enclosed charge is $\rho \cdot \frac{4}{3}\pi r^3 = Q\,r^3/a^3$, so $E \cdot 4\pi r^2 = Qr^3/(\varepsilon_0 a^3)$, giving $E = \dfrac{Qr}{4\pi\varepsilon_0 a^3}$.
- Outside ($r \ge a$): the enclosed charge is Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$.

Step 2: the potential inside

Integrate in from infinity, splitting the range at $r' = a$ because E has different forms on each side:

$$V(r) = \int_r^\infty E\,\mathrm{d}r' = \underbrace{\int_a^\infty \frac{Q}{4\pi\varepsilon_0 r'^2}\,\mathrm{d}r'}_{=\,Q/(4\pi\varepsilon_0 a)} + \int_r^a \frac{Qr'}{4\pi\varepsilon_0 a^3}\,\mathrm{d}r' = \frac{Q}{4\pi\varepsilon_0 a} + \frac{Q}{4\pi\varepsilon_0 a^3}\cdot\frac{a^2 - r^2}{2}.$$

Over the common denominator $8\pi\varepsilon_0 a^3$:

$$V(r) = \frac{2Qa^2 + Q(a^2 - r^2)}{8\pi\varepsilon_0 a^3} = \frac{Q(3a^2 - r^2)}{8\pi\varepsilon_0 a^3} \qquad (r \le a).$$

Checks: $V(a) = 2Qa^2/(8\pi\varepsilon_0 a^3) = Q/(4\pi\varepsilon_0 a)$, continuous with the outside; and $V(0) = \frac{3}{2}V(a)$, the maximum at the centre.

Step 3: the energy integral

ρ is zero outside the sphere, so the integral runs only over $r \le a$. Using shells, $\mathrm{d}\tau = 4\pi r^2\,\mathrm{d}r$:

$$U = \frac{1}{2}\int_0^a \rho\,V(r)\,4\pi r^2\,\mathrm{d}r = \frac{1}{2}\cdot\frac{3Q}{4\pi a^3}\cdot\frac{Q}{8\pi\varepsilon_0 a^3}\cdot 4\pi\int_0^a (3a^2 - r^2)\,r^2\,\mathrm{d}r.$$

The constants simplify to $\dfrac{3Q^2}{16\pi\varepsilon_0 a^6}$, and the integral is

$$\int_0^a (3a^2r^2 - r^4)\,\mathrm{d}r = \left[a^2r^3 - \frac{r^5}{5}\right]_0^a = a^5 - \frac{a^5}{5} = \frac{4a^5}{5}.$$

So

$$U = \frac{3Q^2}{16\pi\varepsilon_0 a^6}\cdot\frac{4a^5}{5} = \frac{12Q^2}{80\pi\varepsilon_0 a} \quad\Longrightarrow\quad \boxed{U = \frac{3Q^2}{20\pi\varepsilon_0 a} = \frac{3}{5}\,\frac{Q^2}{4\pi\varepsilon_0 a}}$$

Cross-check via field energy, $U = \frac{\varepsilon_0}{2}\int E^2\,\mathrm{d}\tau$ over *all* space:

$$\text{inside: } \frac{\varepsilon_0}{2}\int_0^a \left(\frac{Qr}{4\pi\varepsilon_0 a^3}\right)^2 4\pi r^2\,\mathrm{d}r = \frac{Q^2}{8\pi\varepsilon_0 a^6}\int_0^a r^4\,\mathrm{d}r = \frac{Q^2}{40\pi\varepsilon_0 a},$$

$$\text{outside: } \frac{\varepsilon_0}{2}\int_a^\infty \left(\frac{Q}{4\pi\varepsilon_0 r^2}\right)^2 4\pi r^2\,\mathrm{d}r = \frac{Q^2}{8\pi\varepsilon_0}\int_a^\infty \frac{\mathrm{d}r}{r^2} = \frac{Q^2}{8\pi\varepsilon_0 a} = \frac{5Q^2}{40\pi\varepsilon_0 a}.$$

The sum is $6Q^2/(40\pi\varepsilon_0 a) = 3Q^2/(20\pi\varepsilon_0 a)$, as before. Five-sixths of the energy is stored outside the sphere.

Assembling shell by shell. Build the ball up from nothing. When it has radius r it holds $q(r) = Qr^3/a^3$, and the potential at its surface is $q/(4\pi\varepsilon_0 r)$. Bringing in the next shell, $\mathrm{d}q = \rho\, 4\pi r^2\, \mathrm{d}r = 3Qr^2\, \mathrm{d}r/a^3$, from infinity costs

$$\mathrm{d}U = \frac{q(r)\,\mathrm{d}q}{4\pi\varepsilon_0 r} = \frac{1}{4\pi\varepsilon_0 r}\cdot\frac{Qr^3}{a^3}\cdot\frac{3Qr^2\,\mathrm{d}r}{a^3} = \frac{3Q^2 r^4\,\mathrm{d}r}{4\pi\varepsilon_0 a^6}, \qquad U = \frac{3Q^2}{4\pi\varepsilon_0 a^6}\cdot\frac{a^5}{5} = \frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 a}.$$

No factor $\frac{1}{2}$ is needed here, because each piece of charge is moved only once, against the charge already present. This is the clearest reading of the $\frac{1}{2}$ in $\frac{1}{2}\int \rho V\,\mathrm{d}\tau$: it avoids counting each pair twice.

Step 4: the "classical radius" of the electron

Set $Q = e$ and $U = m_e c^2$ (e the elementary charge, m_e the electron mass, c the speed of light), and solve for a:

$$a = \frac{3}{5}\,\frac{e^2}{4\pi\varepsilon_0\, m_e c^2}.$$

Numerically, $e^2/(4\pi\varepsilon_0) = (1.602\times10^{-19})^2 \times 8.988\times10^9 = 2.307\times10^{-28}\,\mathrm{J\,m}$ and $m_e c^2 = (9.109\times10^{-31})(2.998\times10^8)^2 = 8.187\times10^{-14}\,\mathrm{J}$, so

$$a = 0.6\times\frac{2.307\times10^{-28}\,\mathrm{J\,m}}{8.187\times10^{-14}\,\mathrm{J}} = 0.6\times2.818\times10^{-15}\,\mathrm{m} = 1.69\times10^{-15}\,\mathrm{m}.$$

Why the assumption is "dubious"

- **The prefactor is model-dependent:** a uniform ball gives $3/5$, a thin shell $1/2$; the conventional $r_e = e^2/(4\pi\varepsilon_0 m_e c^2) = 2.82\,\mathrm{fm}$ drops it. The calculation fixes a length scale, not a size.
- **The electron is point-like** to at least $\sim 10^{-18}\,\mathrm{m}$ in scattering experiments; for a point charge ($a \to 0$) $U \to \infty$.
- **The model is unstable:** a ball of like charge needs non-electromagnetic ("Poincaré") stresses to hold it together, and purely electromagnetic mass gives the wrong energy–momentum relation (the "4/3 problem").
- **The resolution is quantum:** in quantum electrodynamics the divergent self-energy is absorbed into the measured mass by renormalisation. The length r_e survives in, e.g., the Thomson cross-section $\sigma_T = \frac{8\pi}{3}r_e^2$, marking where classical electrodynamics fails for an electron.

For scale, $a = 1.69\,\mathrm{fm}$ is about twice the proton charge radius (0.84 fm).

Extensions

1. Repeat for a charged spherical shell and explain why $3/5$ becomes $1/2$.
2. Use the result to estimate the electrostatic contribution to the mass difference between mirror nuclei (e.g. ^{3}He and ^{3}H) and compare with experiment.
3. Explain why the energy of a point charge diverges, while the energy of a pair of point charges, excluding self-energies, is finite.

Problem 6 — Conductors and Gauss's law

Problem 6 (A) (as set on the sheet)

Use Gauss' Law to show the following:

(a) Any excess charge placed on a conductor must lie entirely on its surface.

(b) The electric field at the surface of a conductor is normal to the surface and has a magnitude σ/ε_0, where σ is the charge density (per unit area) on the surface.

(c) A closed, hollow conductor shields its interior from fields due to charges outside, but does not shield its exterior from fields due to charges placed inside it.

[The electric field inside a conductor is, of course, always zero.]

Supervisor's note

Where the learning is. Rigorous reasoning about conductors: what follows from $\mathbf{E} = 0$ inside, from Gauss's law, and from $\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = 0$. This is the foundation for images (Problems 7–9) and for the uniqueness theorem.

Conceptually tricky. In (b) and (c), Gauss's law alone is *not* sufficient. Gauss shows the cavity wall has zero *net* charge, but cannot rule out $+$ and $-$ patches; that needs the curl condition or uniqueness. Most students write a plausible but incomplete argument here. The asymmetry in (c) (outside shielded from inside only if earthed) is also worth drawing out.

What you need to know before starting

- **Why $\mathbf{E} = 0$ in a conductor in equilibrium:** a conductor contains free charges; any field inside would push them, so they rearrange until the field vanishes. (This takes about the dielectric relaxation time of Problem 15.)
- **Gauss's law**, integral form $\oint_S \mathbf{E} \cdot \mathrm{d}\mathbf{A} = Q_{\text{enc}}/\varepsilon_0$, and differential form $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$.
- **The electrostatic field is conservative:** $\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = 0$ round any closed loop (equivalently $\nabla \times \mathbf{E} = 0$, or $\mathbf{E} = -\nabla V$). This is *not* part of Gauss's law; parts (b) and (c) need it as well, despite the wording of the question.
- **Uniqueness theorem** (optional, used as an alternative in (c)): in a region bounded by surfaces of specified potential, there is only one solution of Laplace's equation $\nabla^2 V = 0$.

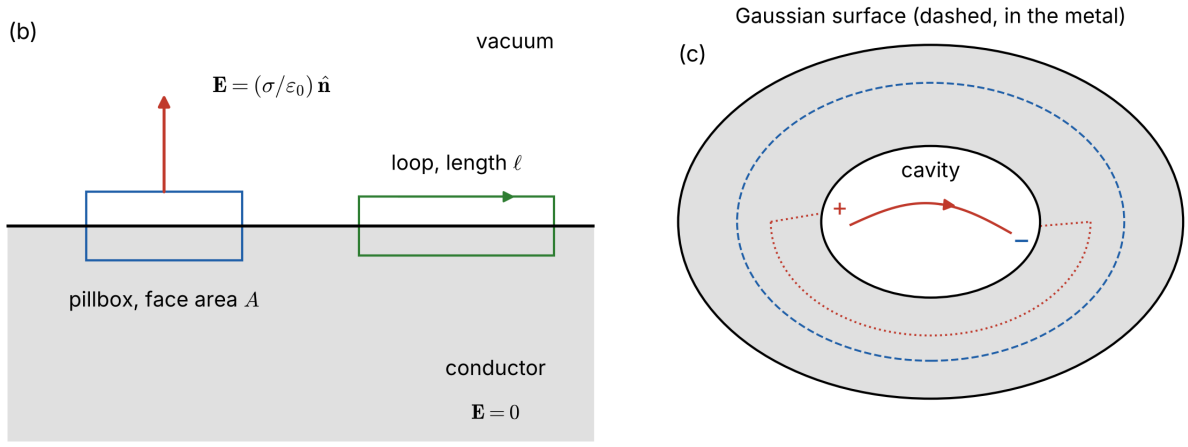


Figure 6: (b) A pillbox (Gauss) and a thin rectangular loop ($\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l}$) straddling a conductor surface. (c) A cavity in a conductor: the dashed Gaussian surface lies in the metal; a field line from a $+$ patch to a $-$ patch, closed through the metal, would give $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l} > 0$.

Starting point. In electrostatic equilibrium $\mathbf{E} = 0$ inside the material of a conductor. Consequently the conductor is an equipotential (since $V(B) - V(A) = -\int_A^B \mathbf{E}\cdot\,\mathrm{d}\mathbf{l} = 0$ for any path through the metal).

(a) Excess charge lies on the surface

Take any closed surface S lying entirely inside the material. $\mathbf{E} = 0$ at every point of S, so

$$Q_{\text{enc}} = \varepsilon_0 \oint_S \mathbf{E}\cdot\,\mathrm{d}\mathbf{A} = 0.$$

S can be shrunk around any interior point, so there is no charge at any interior point: $\rho = 0$ throughout the interior. (In differential form: $\rho = \varepsilon_0 \nabla\cdot\mathbf{E} = \varepsilon_0\nabla\cdot\mathbf{0} = 0$.) Any excess charge must therefore reside on the surface, in practice in a layer about as thick as the screening length ($\sim 0.1\,\text{nm}$ in a metal).

(b) The surface field is normal and equals σ/ε_0

Direction (needs the loop condition). Draw a thin rectangular loop with one long side, of length ℓ, just outside the surface and parallel to it, and the other just inside, in the metal. Let the short sides have length $h \to 0$, so they contribute nothing. Then

$$\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l} = \underbrace{E_t^{\text{out}}\,\ell}_{\text{outside side}} + \underbrace{0\cdot\ell}_{\text{inside side}} + \underbrace{O(h)}_{\text{short sides}\to 0} = 0 \quad\Longrightarrow\quad E_t^{\text{out}} = 0,$$

where E_t is the component of $\mathbf{E}$ tangential to the surface. So just outside, $\mathbf{E}$ has no tangential component: it is normal to the surface. (Equivalently: the surface is an equipotential, and $\mathbf{E} = -\nabla V$ is always perpendicular to equipotentials.)

Magnitude (Gauss's law). Now use a short cylindrical "pillbox" of face area A straddling the surface, small enough that σ is uniform over it. Its flux has three parts:

- outer face: $\mathbf{E} = E_n\hat{\mathbf{n}}$ is parallel to $\mathrm{d}\mathbf{A} = \hat{\mathbf{n}}\,\mathrm{d}A$, contributing $E_n A$;
- inner face: inside the metal, $\mathbf{E} = 0$, contributing 0;

- curved side: its height can be made as small as we like, and $\mathbf{E}$ outside has no component along the side's normal anyway, so it contributes 0.

The pillbox encloses charge σA, so

$$E_n A = \frac{\sigma A}{\varepsilon_0} \quad \Longrightarrow \quad \mathbf{E} = \frac{\sigma}{\varepsilon_0}\,\hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is the outward normal and σ the local surface charge density.

Comparison. An isolated sheet of charge gives $\sigma/(2\varepsilon_0)$ on *each* side. At a conductor the rest of the charge in the universe arranges itself so that it cancels the sheet's field inside the metal and therefore doubles it outside, giving σ/ε_0. (This is also why the force per unit area on the surface is $\sigma^2/2\varepsilon_0$ and not σ^2/ε_0; see Problem 11.)

(c) Shielding

Interior shielded from outside charges. Consider an empty cavity inside a conductor, with arbitrary charges outside.

1. A Gaussian surface lying within the metal and enclosing the cavity has $\mathbf{E} = 0$ everywhere on it, so the total charge on the cavity wall is zero.
2. Gauss's law alone still allows equal amounts of $+$ and $-$ charge on different patches of the wall. Suppose there were such patches. A field line would leave a $+$ patch, cross the cavity, and end on a $-$ patch. Take the closed loop that follows this field line across the cavity and returns through the metal. Along the field line $\mathbf{E}\cdot\mathrm{d}\mathbf{l} > 0$ at every point; in the metal $\mathbf{E} = 0$. So $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l} > 0$, contradicting the conservative property. Hence there are no patches.
3. So the wall is uncharged everywhere and $\mathbf{E} = 0$ throughout the cavity. (Alternatively: $V =$ const satisfies Laplace's equation in the cavity and the boundary condition that the wall is an equipotential; by uniqueness it is the only solution.)

External charges induce charge only on the outer surface, arranged to cancel the external field everywhere in the metal and the cavity. This is a Faraday cage.

Exterior not shielded from inside charges. Now place a charge q in the cavity.

1. The Gaussian surface in the metal still has $\mathbf{E} = 0$ on it, so it encloses zero net charge: the cavity wall carries $-q$.
2. If the conductor was neutral to begin with, charge conservation puts $+q$ on its outer surface.
3. A Gaussian surface drawn outside the whole conductor encloses $q - q + q = q$, so the field outside cannot be zero.

The outer distribution does not depend on where q sits inside: the metal shields the outer surface from the details of the inside, so for a spherical outer surface the charge $+q$ spreads uniformly and the external field is exactly $q/(4\pi\varepsilon_0 r^2)$. **Earthing** the conductor lets the $+q$ drain away and removes the external field, which is why screened enclosures and coaxial cables are earthed.

Extensions

1. Use the uniqueness theorem to show that the field inside an empty cavity of any shape in a conductor is zero.
2. A point charge sits off-centre inside a neutral conducting spherical shell. Sketch the induced charge on the inner and outer surfaces and the field outside.

3. Why does a car or aircraft protect its occupants from lightning even though it is not a closed conductor?

Problem 7 — A small conducting sphere between two charges

Problem 7 (A) (as set on the sheet)

Two point charges q and $-q$ are a distance $2d$ apart. Show that, if a conducting sphere of small radius a ($a \ll 2d$) is placed midway between the two charges, the force on each is increased by a factor of approximately $(1 + 16(a^3/d^3))$. *Hint: you may use what you know about the electric field a long distance away from a dipole.*

Supervisor's note

Where the learning is. Polarisation of a conductor: a sphere in a uniform field becomes a dipole $p = 4\pi\varepsilon_0 a^3 E_0$. Students learn to solve Laplace's equation by matching a trial solution to boundary conditions, and to use the dipole far field in a force calculation.

Conceptually tricky. Students who have not met the induced-dipole result will be stuck; the hint points to the dipole field but not to how p arises. The approximation needs justifying (uniform field over the sphere), and the result is counter-intuitive: a conductor placed between two charges *increases* the force, which conflicts with a naive notion of "shielding".

What you need to know before starting

- **Additional knowledge needed:** the field of a conducting sphere placed in a uniform field. This is *not* given in the question; it is derived in Step 2 by solving Laplace's equation, and checked by the method of images in the Remarks. Students who have not met either technique should be told the result $p = 4\pi\varepsilon_0 a^3 E_0$ and asked to justify it afterwards.
- **Laplace's equation** $\nabla^2 V = 0$ holds wherever there is no charge. Two solutions we need: $V = -E_0 z = -E_0 r\cos\theta$ (a uniform field $E_0\hat{\mathbf{z}}$) and $V = p\cos\theta/(4\pi\varepsilon_0 r^2)$ (a point dipole, Problem 3).
- **Uniqueness theorem:** if a trial potential satisfies Laplace's equation and all the boundary conditions, it is *the* solution.
- **Dipole field on its axis:** $2p/(4\pi\varepsilon_0 r^3)$, directed along $\mathbf{p}$ (Problem 3).

Final answer

The sphere acquires an induced dipole $p = 4\pi\varepsilon_0 a^3 E_0$, whose field at each charge adds an attractive force; the force increases by the factor $1 + 16a^3/d^3$. (Exactly, by images: $1 + 16(a/d)^3/[1-(a/d)^4]^2$.)

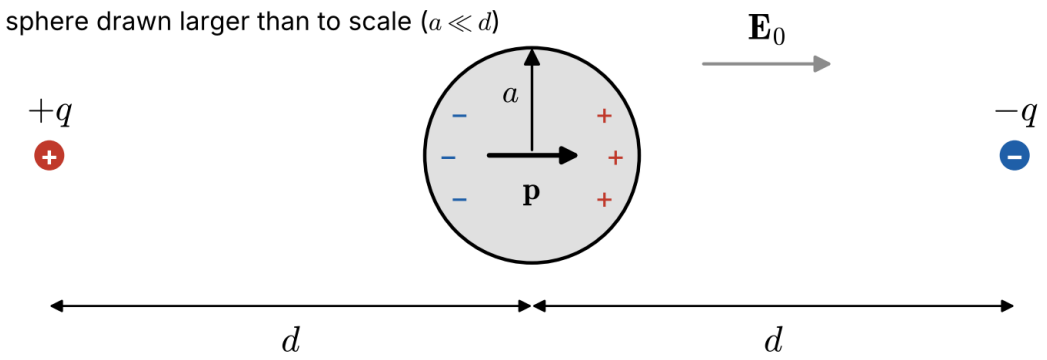


Figure 7: Neutral conducting sphere of radius a midway between $+q$ and $-q$. The applied field $\mathbf{E}_0$ separates its surface charge, creating the induced dipole $\mathbf{p}$.

Set-up. $+q$ at $x = -d$, $-q$ at $x = +d$, and an uncharged, isolated conducting sphere of radius $a \ll d$ at the origin. Without the sphere the two charges attract with force $F_0 = q^2/[4\pi\varepsilon_0(2d)^2]$.

Step 1: the applied field at the sphere

At the origin, the field of $+q$ points away from it, i.e. along $+x$; the field of $-q$ points towards it, also along $+x$. Each has magnitude $q/(4\pi\varepsilon_0 d^2)$, so

$$\mathbf{E}_0 = \frac{2q}{4\pi\varepsilon_0 d^2}\,\hat{\mathbf{x}}.$$

Is it uniform over the sphere? On the x-axis, $E_x(x) = \dfrac{q}{4\pi\varepsilon_0}\left[\dfrac{1}{(d+x)^2} + \dfrac{1}{(d-x)^2}\right]$, which is even in x, so its first derivative vanishes at the centre and it changes only at second order, by a relative amount $\sim 3x^2/d^2 \lesssim 3a^2/d^2$. We will see below that this non-uniformity does not change the induced dipole at all; it only adds much smaller higher multipoles.

Step 2: a conducting sphere in a uniform field

Use spherical polars centred on the sphere, with the polar axis along $\mathbf{E}_0$ (so here θ is measured from $+x$). We need $V(r,\theta)$ outside the sphere that

(i) satisfies Laplace's equation for $r > a$ (no charge there),

(ii) tends to the applied potential $-E_0 r\cos\theta$ far away, and

(iii) is constant on the sphere $r = a$ (a conductor is an equipotential).

Trial solution: the applied field plus a dipole at the centre,

$$V(r,\theta) = -E_0 r\cos\theta + \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}.$$

Both terms solve Laplace's equation, and the dipole term dies away as $r \to \infty$, so (i) and (ii) hold. For (iii): the sphere is neutral and the set-up is antisymmetric, so its potential is the same as that of the mid-plane, $V = 0$. Setting $r = a$:

$$V(a,\theta) = \cos\theta\left(-E_0 a + \frac{p}{4\pi\varepsilon_0 a^2}\right) = 0 \text{ for all } \theta \quad\Longrightarrow\quad \boxed{p = 4\pi\varepsilon_0 a^3 E_0}$$

By uniqueness this is *the* solution: outside the sphere, the induced surface charge produces exactly the field of a point dipole $\mathbf{p}$ at the centre, pointing along $\mathbf{E}_0$. The surface charge is

$\sigma = \varepsilon_0 E_r(a) = -\varepsilon_0\,\partial V/\partial r|_a = \varepsilon_0(E_0 + 2E_0)\cos\theta = 3\varepsilon_0 E_0\cos\theta$: negative on the side facing $+q$ ($\theta = \pi$), positive on the side facing $-q$.

Step 3: field of the induced dipole at each charge

$+q$ lies on the dipole's axis at distance d (on the side opposite to where $\mathbf{p}$ points). The on-axis dipole field is along $\mathbf{p}$ on both sides:

$$E_{\text{dip}} = \frac{2p}{4\pi\varepsilon_0 d^3} = \frac{2\cdot 4\pi\varepsilon_0 a^3 E_0}{4\pi\varepsilon_0 d^3} = \frac{2a^3}{d^3}\cdot\frac{2q}{4\pi\varepsilon_0 d^2} = \frac{4qa^3}{4\pi\varepsilon_0 d^5},$$

pointing along $+x$, i.e. towards $-q$. The force qE_{dip} on $+q$ therefore adds to the attraction. By symmetry the same holds for $-q$.

Step 4: ratio of forces

$$F_0 = \frac{q^2}{4\pi\varepsilon_0(2d)^2} = \frac{q^2}{16\pi\varepsilon_0 d^2}, \qquad \Delta F = qE_{\text{dip}} = \frac{4q^2a^3}{4\pi\varepsilon_0 d^5} = \frac{q^2a^3}{\pi\varepsilon_0 d^5},$$

$$\frac{F_0 + \Delta F}{F_0} = 1 + \frac{q^2a^3/(\pi\varepsilon_0 d^5)}{q^2/(16\pi\varepsilon_0 d^2)} = 1 + 16\,\frac{a^3}{d^3}.$$

Remarks

- **Picture:** the induced $-$ charge sits nearer $+q$ than $-q$ does (and vice versa), so the attraction grows. A conductor between two charges that encloses neither enhances, rather than screens, their interaction.
- **Exact check by images.** A point charge Q at distance D from the centre of an earthed sphere has an image $-Qa/D$ at distance a^2/D from the centre, on the same side (the standard sphere image). Here $+q$ at $x = -d$ has image $-qa/d$ at $x = -a^2/d$, and $-q$ at $x = +d$ has image $+qa/d$ at $x = +a^2/d$. These images have zero net charge, so the sphere stays neutral with no extra charge at its centre, and they form a dipole $(qa/d)(2a^2/d) = 2qa^3/d^2 = 4\pi\varepsilon_0 a^3 E_0$, as in Step 2. The extra force on $+q$ from the two images, with $\epsilon \equiv a^2/d^2$, is

  $$\Delta F = \frac{q^2(a/d)}{4\pi\varepsilon_0 d^2}\left[\frac{1}{(1-\epsilon)^2} - \frac{1}{(1+\epsilon)^2}\right] = \frac{q^2(a/d)}{4\pi\varepsilon_0 d^2}\cdot\frac{4\epsilon}{(1-\epsilon^2)^2}, \quad \text{so} \quad \frac{F}{F_0} = 1 + \frac{16(a/d)^3}{[1-(a/d)^4]^2}.$$

  The leading correction is therefore of relative size $2(a/d)^4$, even smaller than the $(a/d)^2$ one might guess from the non-uniformity of $\mathbf{E}_0$.
- **Polarisability:** $\alpha = p/E_0 = 4\pi\varepsilon_0 a^3$ scales with volume; for $a \sim 0.1\,\text{nm}$ it gives the order of atomic polarisabilities, $\sim 4\pi\varepsilon_0 \times 10^{-30}\,\text{m}^3$.
- **Why 16:** $16 = 2\times 2\times 4$ — two charges contribute to E_0, the axial dipole field carries a factor 2, and the original separation is $2d$, which puts 4 in F_0. Even $a = 0.1d$ raises the force by 1.6%.

Extensions

1. Use the full image solution to find the next correction, and estimate how small a/d must be for 1% accuracy.
2. Repeat with the sphere earthed rather than isolated: what changes, and why?
3. Replace the conductor by a dielectric sphere of relative permittivity ϵ and show the factor becomes $1 + 16\,\frac{\epsilon-1}{\epsilon+2}\,\frac{a^3}{d^3}$.

Problem 8 — Thundercloud above conducting ground

Problem 8 (A) (as set on the sheet)

The electrical system of a typical thundercloud can be represented by a vertical dipole consisting of a charge of $+40\,\mathrm{C}$ at a height of $10\,\mathrm{km}$ and a charge of $-40\,\mathrm{C}$ vertically below it at a height of $6\,\mathrm{km}$. What is the electric field at the ground immediately below the thundercloud and at what distance from there is the field at the ground zero? *Hint: treat the ground as a perfect conductor.*

Supervisor's note

Where the learning is. The method of images for a plane, and the superposition of image pairs. Physically, it shows how a thundercloud reverses the fair-weather field and why the ground under a storm carries positive charge.

Conceptually tricky. Students frequently get the direction of the field at the ground wrong, or forget the factor of 2 from the image. Locating the zero requires an algebraic step (a 3/2-power ratio) that many will find awkward. Ask why the far field falls as ρ^{-3}: the cloud's net dipole moment is non-zero.

What you need to know before starting

- **Method of images for a plane:** the field above an earthed conducting plane due to a charge Q at height h is the same as the field of Q together with an "image" charge $-Q$ at the mirror point a depth h below the plane. Reason: the pair makes $V = 0$ everywhere on the plane, and by the **uniqueness theorem** the field above the plane is fixed by the charges there plus the boundary value $V = 0$.
- The image charges are a calculational device. Physically, the field below ground is zero; the images stand in for the charge induced on the ground's surface.
- Problem 6(b): the field at a conductor is normal to it, with $\sigma = \varepsilon_0 E_n$.

Final answer

Directly below the cloud $E = 12.8\,\mathrm{kV\,m^{-1}}$, pointing upwards; the ground field vanishes on a circle of radius $11.0\,\mathrm{km}$ around that point.

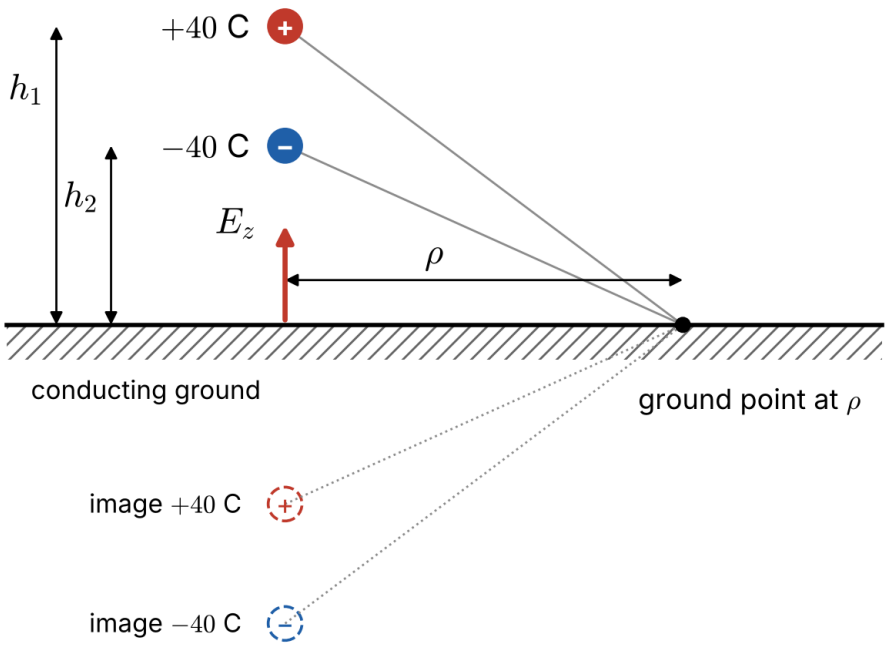


Figure 8: The cloud charges at heights h_1 and h_2, their images below the conducting ground, and a ground point at horizontal distance ρ. E_z is the vertical field at the ground.

Set-up. The ground is the conducting plane $z = 0$, with z measured upwards. $Q_1 = 40\,\mathrm{C}$ at $h_1 = 10\,\mathrm{km}$, $Q_2 = -40\,\mathrm{C}$ at $h_2 = 6\,\mathrm{km}$; ρ is the horizontal distance of a ground point from the point directly beneath the cloud.

Step 1: method of images

Charge	Real or image	Height z
$+40$ C	real	$+10$ km
-40 C	real	$+6$ km
$+40$ C	image of the -40 C	-6 km
-40 C	image of the $+40$ C	-10 km

Every point of $z = 0$ is equidistant from each charge and its oppositely signed image, so their potentials cancel and $V = 0$ there. By uniqueness this four-charge field is the actual field above ground.

Step 2: one charge and its image

Take a charge Q at $(0, 0, h)$ and its image $-Q$ at $(0, 0, -h)$, and the ground point $(\rho, 0, 0)$. The vectors *from* the charges *to* the ground point are

$$\mathbf{r}_Q = (\rho, 0, -h), \qquad \mathbf{r}_{-Q} = (\rho, 0, +h), \qquad |\mathbf{r}_Q| = |\mathbf{r}_{-Q}| = R = \sqrt{\rho^2 + h^2}.$$

By Coulomb's law,

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_0 R^3}(\rho, 0, -h) + \frac{-Q}{4\pi\varepsilon_0 R^3}(\rho, 0, h) = \frac{Q}{4\pi\varepsilon_0 R^3}(0,\, 0,\, -2h),$$

so

$$E_z(\rho) = -\frac{2Qh}{4\pi\varepsilon_0(\rho^2+h^2)^{3/2}}.$$

The horizontal components cancel, so the field is normal to the ground, as Problem 6(b) requires. The image doubles the vertical component. For $Q > 0$ the field points down.

Step 3: superpose the two pairs

$$E_z(\rho) = -\frac{2}{4\pi\varepsilon_0}\left[\frac{Q_1 h_1}{(\rho^2+h_1^2)^{3/2}} + \frac{Q_2 h_2}{(\rho^2+h_2^2)^{3/2}}\right].$$

Directly beneath the cloud, $\rho = 0$, and $h/(h^2)^{3/2} = 1/h^2$:

$$\begin{aligned} E_z(0) &= -\frac{2}{4\pi\varepsilon_0}\left[\frac{Q_1}{h_1^2} + \frac{Q_2}{h_2^2}\right] = -(1.798\times10^{10}\,\mathrm{m\,F^{-1}})\left[\frac{40\,\mathrm{C}}{(1.0\times10^4\,\mathrm{m})^2} - \frac{40\,\mathrm{C}}{(6.0\times10^3\,\mathrm{m})^2}\right] \\ &= -(1.798\times10^{10})\left(4.000\times10^{-7} - 1.111\times10^{-6}\right)\,\mathrm{V\,m^{-1}} = 1.28\times10^4\,\mathrm{V\,m^{-1}}, \end{aligned}$$

i.e. $12.8\,\mathrm{kV\,m^{-1}}$ **pointing upwards**. The nearer negative charge dominates. The ground beneath carries positive induced charge $\sigma = \varepsilon_0 E_z \approx 1.1\times10^{-7}\,\mathrm{C\,m^{-2}}$.

Step 4: where the ground field vanishes

Set the bracket to zero. Since $Q_2 = -Q_1$, this requires

$$\frac{h_1}{(\rho^2+h_1^2)^{3/2}} = \frac{h_2}{(\rho^2+h_2^2)^{3/2}} \quad\Longleftrightarrow\quad \left(\frac{\rho^2+h_1^2}{\rho^2+h_2^2}\right)^{3/2} = \frac{h_1}{h_2}.$$

Raise both sides to the power $2/3$ to remove the $3/2$ power. With lengths in km,

$$\frac{\rho^2+100}{\rho^2+36} = \left(\frac{10}{6}\right)^{2/3} = 1.4057.$$

Multiply out and collect ρ^2:

$$\rho^2 + 100 = 1.4057\,\rho^2 + 50.61 \quad\Longrightarrow\quad 0.4057\,\rho^2 = 49.39,$$

$$\rho^2 = 121.7\,\mathrm{km^2}, \qquad \rho = 11.0\,\mathrm{km}.$$

By the cylindrical symmetry about the vertical through the cloud, the zero lies on a circle of this radius (numerical root: 11.03 km).

Interpretation

- **Two regimes:** inside the circle the nearer negative charge wins and E points up; outside, the higher positive charge wins and E points down, like the fair-weather field of Problem 2(b).
- **Far field:** for $\rho \gg h$, $(\rho^2+h^2)^{3/2} \approx \rho^3$ and the bracket tends to $(Q_1h_1 + Q_2h_2)/\rho^3$. The cloud is a vertical dipole of moment $Q_1h_1 + Q_2h_2 = 40\,\mathrm{C}\times4\,\mathrm{km} = 1.6\times10^5\,\mathrm{C\,m}$, doubled by its image, so $E_z \propto \rho^{-3}$: at 50 km it is $2\times8.988\times10^9\times1.6\times10^5/(5\times10^4)^3 \approx 23\,\mathrm{V\,m^{-1}}$ downwards.
- **Real storms:** $12.8\,\mathrm{kV\,m^{-1}}$ is about 130 times the fair-weather field; sharp objects concentrate it further, launching corona discharge (St Elmo's fire) and upward leaders.
- **The approximation:** damp ground is a good conductor on the storm's timescale because its dielectric relaxation time (Problem 15) is far shorter than the minutes over which cloud charge builds up.

Extensions

1. Find the total induced ground charge (why is it zero here?) and sketch $\sigma(\rho)$.
2. Estimate the potential difference between the head and feet of a person 1.8 m tall standing directly under the cloud.
3. Real clouds also carry a small positive charge at their base. Add $+10\,\mathrm{C}$ at 2 km and recompute the field directly below.

Problem 9 — A charge inside a 60° conducting wedge

Problem 9 (A) (as set on the sheet)

Two infinite conducting plane sheets meet at an angle of 60°. A particle carrying a charge q is constrained to move on the plane bisecting this angle. Obtain an expression for the force on the particle when it is a distance x from the line on which the sheets meet.

Supervisor's note

Where the learning is. Images beyond the single plane: constructing the full image set by repeated reflection, verifying the boundary conditions, and exploiting symmetry to reduce the force to one component. This is the most conceptually demanding problem in the first ten.

Conceptually tricky. Many students include only the two direct reflections and miss the three higher-order images. Others place images inside the physical region. The question of *why* the method works only for wedge angles $180^\circ/n$ is the deepest point, and worth a few minutes of discussion. Also watch for students wrongly halving the force.

What you need to know before starting

- **Method of images for a plane** (Problem 8): a conducting plane at $V = 0$ can be replaced by an opposite charge at the mirror point. With two sheets, the images must themselves be mirrored in both sheets until the pattern closes.
- **Chord length:** two points at distance x from the origin and separated by angle Δ are $|\mathbf{r} - \mathbf{r}'| = \sqrt{x^2 + x^2 - 2x^2\cos\Delta} = 2x\sin(\Delta/2)$ apart (cosine rule, then $1 - \cos\Delta = 2\sin^2(\Delta/2)$).
- The particle is *constrained* to the bisecting plane, so only the force component along the bisector would move it; by symmetry that is in fact the whole force.

Final answer

The force points towards the edge where the sheets meet, with magnitude $F = \left(5 - \dfrac{4}{\sqrt{3}}\right)\dfrac{q^2}{16\pi\varepsilon_0 x^2} \approx 2.69\,\dfrac{q^2}{16\pi\varepsilon_0 x^2}$.

Step 1: the images

Work in the plane perpendicular to the edge, with the edge at the origin. The sheets are the lines at 0° and 60°, and the charge is on the bisector, at distance x from the edge and angle 30°.

The image pattern. Put six charges on the circle of radius x, one every 60° starting from the real

charge, with alternating signs:

$$+q \text{ at } 30^\circ \text{ (real)}, \quad -q \text{ at } 90^\circ, \quad +q \text{ at } 150^\circ, \quad -q \text{ at } 210^\circ, \quad +q \text{ at } 270^\circ, \quad -q \text{ at } 330^\circ.$$

(Mirroring the real charge in the two sheets gives $-q$ at $-30^\circ = 330^\circ$ and at 90°; mirroring those again gives the $+q$ at 150° and 270°, and then the $-q$ at 210°. The pattern closes after six charges because $360^\circ/60^\circ = 6$ is even.)

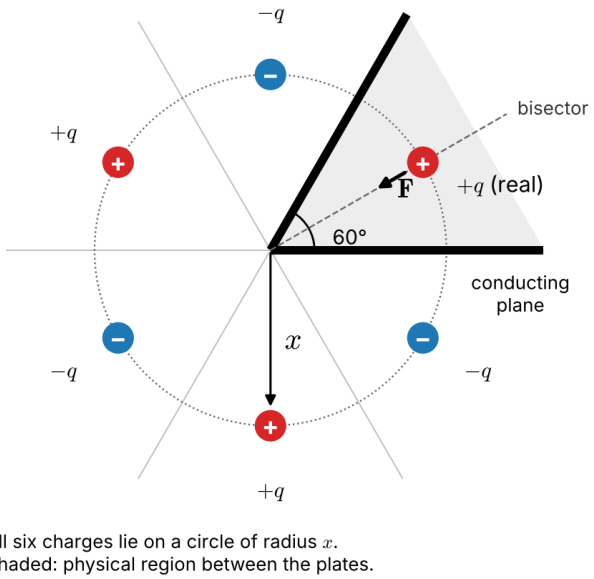


Figure 9: The real charge on the bisector of the 60° wedge and its five images, all on a circle of radius x.

Check that both sheets are at $V = 0$. Each sheet is a mirror line of the pattern that swaps $+q$ with $-q$. For the sheet at 0° the mirror pairs are $30^\circ \leftrightarrow 330^\circ$, $90^\circ \leftrightarrow 270^\circ$ and $150^\circ \leftrightarrow 210^\circ$; for the sheet at 60° they are $30^\circ \leftrightarrow 90^\circ$, $150^\circ \leftrightarrow 330^\circ$ and $210^\circ \leftrightarrow 270^\circ$. Every point of a sheet is equally far from the two members of each pair, so their potentials cancel there. Only the real charge lies inside the wedge, so by uniqueness this is the true field in the wedge.

Step 2: force on the real charge

Separations from the real charge (chord formula, Δ = angular separation):

Angle ϕ	Charge	Δ	Distance $2x\sin(\Delta/2)$	$1-\cos\Delta$
90°, 330°	$-q$	60°	x	$1/2$
150°, 270°	$+q$	120°	$\sqrt{3}\,x$	$3/2$
210°	$-q$	180°	$2x$	2

Which component survives. The images come in pairs placed symmetrically about the bisector ($30^\circ \pm 60^\circ$ and $30^\circ \pm 120^\circ$), plus one on the bisector line itself (210°). So the components of force perpendicular to the bisector cancel, and only the component along $\hat{\mathbf{r}}$, the unit vector from the edge to the real charge, remains.

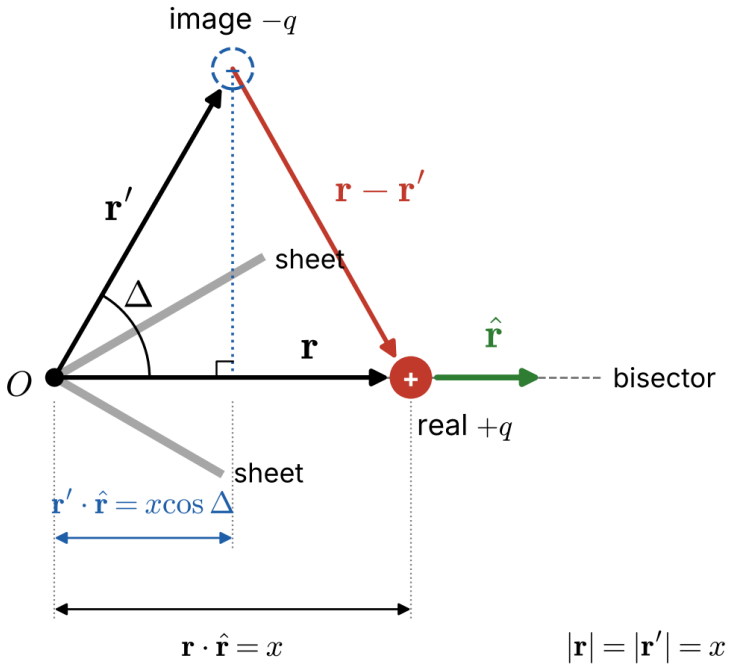


Figure 10: The vectors used in Step 2, drawn with the bisector horizontal and for the image $-q$ at angular separation $\Delta = 60°$ (any image works the same way). O is the edge of the wedge; $\mathbf{r}$ and $\mathbf{r}'$ are the position vectors of the real charge and the image, both of length x; $\mathbf{r}-\mathbf{r}'$ points from the image to the real charge (the Coulomb force on the real charge is along $\mathbf{r}-\mathbf{r}'$ for a like image and opposite to it for an unlike one, as here); $\hat{\mathbf{r}}$ is the unit vector along the bisector. The dot products are projections onto the bisector: $\mathbf{r}\cdot\hat{\mathbf{r}} = x$ and $\mathbf{r}'\cdot\hat{\mathbf{r}} = x\cos\Delta$.

That component. The force on q (at $\mathbf{r}$) from a charge q' (at $\mathbf{r}'$) is $\mathbf{F} = \dfrac{qq'}{4\pi\varepsilon_0}\dfrac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}$. Its component along $\hat{\mathbf{r}}$ uses $\mathbf{r}\cdot\hat{\mathbf{r}} = x$ and $\mathbf{r}'\cdot\hat{\mathbf{r}} = x\cos\Delta$ (Figure 10):

$$F_r = \frac{qq'}{4\pi\varepsilon_0}\frac{x(1-\cos\Delta)}{|\mathbf{r}-\mathbf{r}'|^3}.$$

With $F_0 = q^2/(4\pi\varepsilon_0 x^2)$ and $F_r > 0$ meaning away from the edge:

- two $-q$ at $\Delta = 60°$: each $-F_0\cdot\dfrac{1/2}{1^3} = -\frac{1}{2}F_0$; together $-F_0$;
- two $+q$ at $\Delta = 120°$: each $+F_0\cdot\dfrac{3/2}{(\sqrt{3})^3} = +F_0\dfrac{3/2}{3\sqrt{3}} = \dfrac{F_0}{2\sqrt{3}}$; together $+F_0/\sqrt{3}$;
- one $-q$ at $\Delta = 180°$: $-F_0\cdot\dfrac{2}{2^3} = -\frac{1}{4}F_0$.

Adding, and writing $F_0 = 4\times q^2/(16\pi\varepsilon_0 x^2)$:

$$F_r = F_0\left(-1+\frac{1}{\sqrt{3}}-\frac{1}{4}\right) = -\frac{q^2}{16\pi\varepsilon_0 x^2}\left(4-\frac{4}{\sqrt{3}}+1\right) = -\left(5-\frac{4}{\sqrt{3}}\right)\frac{q^2}{16\pi\varepsilon_0 x^2}.$$

The minus sign means the charge is pulled towards the edge. Numerically $5-4/\sqrt{3} = 2.69$.

Remarks

- **Consistency:** the three $-q$ give $1.25F_0$ of attraction, the two $+q$ only $0.577F_0$ of repulsion; attraction must win, since the induced charge is opposite to q and nearest to it.

- **Single plane comparison:** each sheet is at distance $x \sin 30^\circ = x/2$ from the charge, and alone would give $q^2/[4\pi\varepsilon_0(2 \cdot x/2)^2] = 4q^2/(16\pi\varepsilon_0 x^2)$ normal to itself. The two normals are 120° apart, so the naive sum has magnitude 4 (in these units), along the bisector. The true result, 2.69, is smaller: each sheet's induced charge induces charge on the other, which the higher images account for, and their net effect is repulsive.
- **When images fail:** for a wedge angle not equal to $180^\circ/n$ (n an integer) the reflections never close and images would fall inside the physical region; one then needs separation of variables with Bessel functions of non-integer order.
- **No factor of $\frac{1}{2}$ in the force:** the interaction *energy* with induced charge is half that of the full image system, but the force on the real charge is the full Coulomb force of all the images.

Extensions

1. Repeat for a 90° wedge (three images) and compare the force at the same distance from the edge.
2. Find the induced charge density on one sheet as a function of distance from the edge.
3. For a general wedge angle π/n, set up the image sum for the force and evaluate it numerically for $n = 1$ to 6.

Problem 10 — Heat dissipated when two charged spheres are connected

Problem 10 (A) (as set on the sheet)

Two isolated spherical conductors of radii 30 and 90 mm are charged to 1.5 and 3 kV respectively. They are then connected by a fine resistive wire. How much heat will be generated in the wire in total, if its resistance is sufficient to overdamp the system? *Hint: Treat the problem as a reduced circuit of two capacitors and a resistor.*

Supervisor's note

Where the learning is. Energy bookkeeping in circuits: charge conservation sets the final state, and the dissipated energy $\frac{1}{2}C_s(\Delta V)^2$ is independent of the resistance. Mapping a physical system onto a reduced circuit is a useful modelling skill.

Conceptually tricky. The result that R cancels surprises most students, as does the "missing energy" when $R \to 0$. The role of the word "overdamped" (excluding oscillation and radiation) is rarely appreciated. Numerically, subtracting two nearly equal energies is a trap for precision.

What you need to know before starting

- **Capacitance of an isolated sphere** of radius a: $C = 4\pi\varepsilon_0 a$ (Problem 2), and the stored energy $U = \frac{1}{2}CV^2 = Q^2/2C$.
- **Charge conservation:** the wire only moves charge from one sphere to the other.
- **RC circuits:** solving $R\,\mathrm{d}q/\mathrm{d}t =$ (driving voltage) $- q/C$, and Joule heating $P = I^2R$.
- **Additional knowledge (for "overdamped"):** any real loop of wire has some self-inductance L, so the circuit is really a series LCR circuit, which oscillates unless $R > 2\sqrt{L/C}$. This is covered in the inductance part of the course (Sheet 2).

Final answer

$H = 2.8 \times 10^{-6}\,\mathrm{J}$, independent of the wire's resistance.

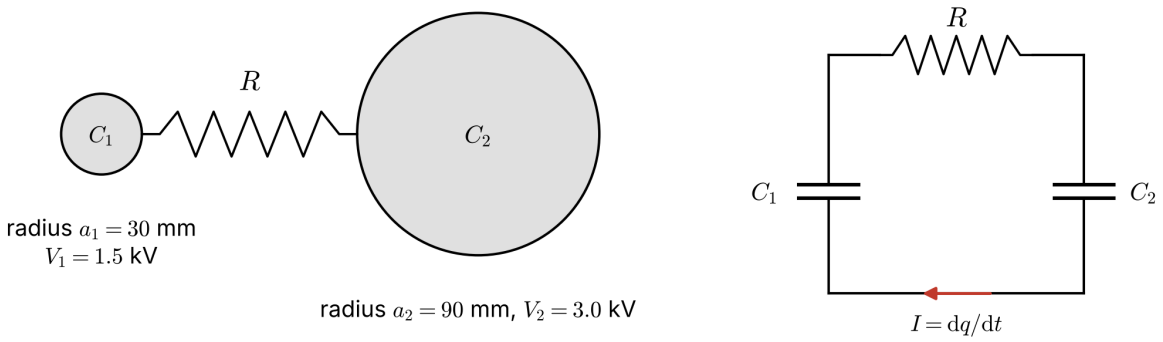


Figure 11: The two spheres joined through a wire of resistance R (left), and the equivalent series circuit (right).

Assumptions. The spheres are far apart compared with their radii (so each one's potential is set by its own charge; their mutual capacitance is negligible), and the wire's own capacitance is negligible.

Step 1: capacitances and charges

With $4\pi\varepsilon_0 = 1.1127 \times 10^{-10}\,\mathrm{F\,m^{-1}}$:

$$C_1 = 4\pi\varepsilon_0(0.030\,\mathrm{m}) = 3.338\,\mathrm{pF}, \qquad C_2 = 4\pi\varepsilon_0(0.090\,\mathrm{m}) = 10.01\,\mathrm{pF} = 3C_1,$$

$$Q_1 = C_1V_1 = (3.338\,\mathrm{pF})(1500\,\mathrm{V}) = 5.007\,\mathrm{nC}, \qquad Q_2 = C_2V_2 = (10.01\,\mathrm{pF})(3000\,\mathrm{V}) = 30.04\,\mathrm{nC}.$$

Step 2: the final state

Charge flows until the two spheres (now one conductor) reach a common potential V_f. The total charge is conserved:

$$C_1V_f + C_2V_f = Q_1 + Q_2 = C_1V_1 + C_2V_2,$$

$$V_f = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = \frac{1 \times 1.5 + 3 \times 3.0}{1 + 3}\,\mathrm{kV} = 2.625\,\mathrm{kV},$$

where we used $C_2 = 3C_1$ and cancelled C_1.

Step 3: energy before and after

$$U_i = \tfrac{1}{2}C_1V_1^2 + \tfrac{1}{2}C_2V_2^2 = 3.755\,\mu\mathrm{J} + 45.06\,\mu\mathrm{J} = 48.82\,\mu\mathrm{J},$$
$$U_f = \tfrac{1}{2}(C_1 + C_2)V_f^2 = \tfrac{1}{2}(13.35\,\mathrm{pF})(2625\,\mathrm{V})^2 = 46.00\,\mu\mathrm{J}.$$

The heat is the difference, $H = U_i - U_f \approx 2.8\,\mu\text{J}$. Subtracting two nearly equal numbers loses precision, so it is better to find a closed form first. Substitute V_f:

$$\begin{aligned} H &= \tfrac{1}{2}C_1V_1^2 + \tfrac{1}{2}C_2V_2^2 - \frac{(C_1V_1 + C_2V_2)^2}{2(C_1+C_2)} \\ &= \frac{(C_1V_1^2 + C_2V_2^2)(C_1+C_2) - (C_1^2V_1^2 + 2C_1C_2V_1V_2 + C_2^2V_2^2)}{2(C_1+C_2)} \\ &= \frac{C_1C_2V_1^2 + C_1C_2V_2^2 - 2C_1C_2V_1V_2}{2(C_1+C_2)} = \frac{1}{2}\,\frac{C_1C_2}{C_1+C_2}\,(V_1-V_2)^2. \end{aligned}$$

(The $C_1^2V_1^2$ and $C_2^2V_2^2$ terms cancel in the second line.) With $C_s = C_1C_2/(C_1+C_2) = 2.503\,\text{pF}$, the two capacitors in series as seen from the wire,

$$H = \tfrac{1}{2}(2.503\,\text{pF})(1500\,\text{V})^2 = 2.82\,\mu\text{J}.$$

Step 4: the reduced circuit, and why R drops out

The loop is C_1, R, C_2 in series. Let $q(t)$ be the charge that has passed through the wire from sphere 2 to sphere 1. Then sphere 2 is at $V_2 - q/C_2$ and sphere 1 at $V_1 + q/C_1$, and the voltage across the wire is

$$\Delta V = \left(V_2 - \frac{q}{C_2}\right) - \left(V_1 + \frac{q}{C_1}\right) = (V_2 - V_1) - \frac{q}{C_s}.$$

Ohm's law, $I = \mathrm{d}q/\mathrm{d}t = \Delta V/R$, gives a first-order equation. Differentiate it once: $\mathrm{d}I/\mathrm{d}t = -I/(RC_s)$, so I decays exponentially from its initial value $(V_2 - V_1)/R$:

$$I(t) = \frac{V_2 - V_1}{R}\,e^{-t/\tau}, \qquad \tau = RC_s.$$

The heat generated is

$$H = \int_0^\infty I^2R\,\mathrm{d}t = \frac{(V_2-V_1)^2}{R}\int_0^\infty e^{-2t/\tau}\,\mathrm{d}t = \frac{(V_2-V_1)^2}{R}\cdot\frac{\tau}{2} = \tfrac{1}{2}C_s(V_2-V_1)^2.$$

R cancels: a larger resistance gives a smaller current for a longer time $\tau = RC_s$, and the total heat is the same.

Why "overdamped"?

With the wire's self-inductance L the circuit is a series LCR circuit with capacitance C_s; it is overdamped when $R > 2\sqrt{L/C_s}$. For smaller R the charge would slosh back and forth between the spheres at $\omega \approx 1/\sqrt{LC_s}$, and an oscillating current radiates some energy away. Overdamping ensures that all 2.8 µJ ends up as Joule heat in the wire.

Remarks

- **The "missing energy":** the loss $\frac{1}{2}C_s(\Delta V)^2$ persists even as $R \to 0$; the energy then leaves as radiation via the oscillations just described. This is the classic paradox of connecting a charged capacitor to an uncharged one, where exactly half the energy is always lost.
- **Charge redistribution:** finally $Q_1' = C_1V_f = 8.76\,\text{nC}$ and $Q_2' = 26.29\,\text{nC}$. Charge divides in proportion to radius, so the surface field $\sigma/\varepsilon_0 = V_f/a$ for a sphere of radius a is three times larger on the smaller sphere — why charge concentrates at sharp points.

Extensions

1. With inductance $L \neq 0$, show that the total heat is unchanged in the underdamped case if radiation is neglected.
2. Estimate the time constant for a $1\,\mathrm{M}\Omega$ wire, and the inductance below which the circuit would oscillate.
3. Repeat for three spheres connected in turn by a single wire. Is the total heat path-dependent?

Problem 11 — Pressure on a charged conducting surface

Problem 11 (A) (as set on the sheet)

Derive the expression for the pressure on a charged surface of a conductor by considering the energy of a charged parallel-plate capacitor when the capacitor is kept (a) at constant charge, (b) at constant voltage. *Hint: ensure that you distinguish between the force acting between the plates, and the external force opposing it.*

Supervisor's note

Where the learning is. Using energy to find forces, and the virtual-work argument with the correct bookkeeping of *all* energy sources (field energy, battery work, mechanical work). The result $p = \sigma^2/2\varepsilon_0 = \frac{1}{2}\varepsilon_0 E^2$ introduces electrostatic pressure, or "field stress".

Conceptually tricky. The constant-voltage case: naively writing $F = -\mathrm{d}U/\mathrm{d}x$ gives a *repulsive* force, because it ignores the work done by the battery, which is twice the change in field energy. A second subtlety is that the force per unit area is $\sigma \times \sigma/(2\varepsilon_0)$, not $\sigma \times \sigma/\varepsilon_0$: a surface charge feels the *average* of the fields just inside (0) and just outside (σ/ε_0), since it does not act on itself.

What you need to know before starting

- **Parallel-plate capacitor** (edges neglected): field $E = \sigma/\varepsilon_0 = V/x$ between plates of area A a distance x apart, capacitance $C = \varepsilon_0 A/x$, stored energy $U = Q^2/2C = \frac{1}{2}CV^2 = \frac{1}{2}\varepsilon_0 E^2 \times (\text{volume } Ax)$.
- **Virtual work.** Imagine moving one plate slowly by $\mathrm{d}x$. Energy conservation: (work done on the system by the external agent) + (work done by any battery) = (change in stored energy). No kinetic energy is produced because the move is quasi-static.
- **Battery work:** a battery of e.m.f. V that pushes charge $\mathrm{d}Q$ round the circuit does work $V\,\mathrm{d}Q$.

Final answer

In both cases the attractive force per unit area on each plate is $p = \dfrac{\sigma^2}{2\varepsilon_0} = \frac{1}{2}\varepsilon_0 E^2$, where σ is the surface charge density and $E = \sigma/\varepsilon_0$ the field at the surface.

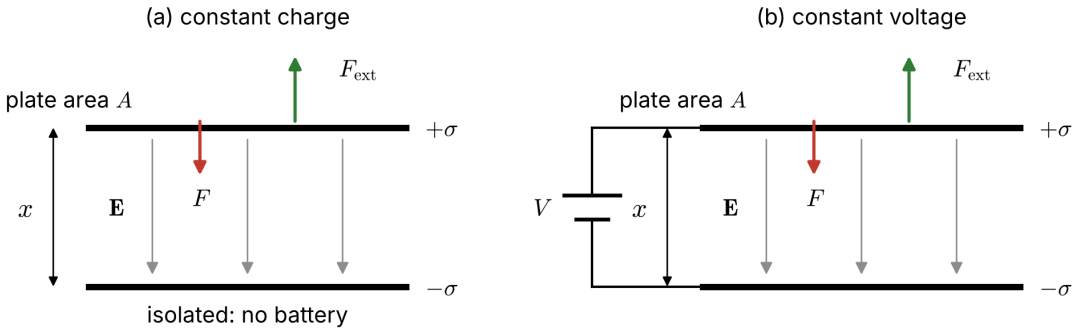


Figure 12: Virtual work on parallel plates a distance x apart: (a) isolated plates, constant charge; (b) plates held at potential difference V by a battery. F is the electric force on the upper plate, F_{ext} the balancing external force.

Set-up. Parallel plates of area A at separation x, with edge effects neglected, so $C = \varepsilon_0 A/x$. The plates carry $\pm Q$ and attract each other with a force of magnitude F (the *electric* force). To hold the upper plate still, an external agent pulls it with an equal and opposite force $F_{\text{ext}} = F$ (the hint's distinction). Now let the agent move the plate apart by $\mathrm{d}x$, slowly: it does work $F_{\text{ext}}\,\mathrm{d}x = F\,\mathrm{d}x$ on the system.

(a) Constant charge Q (isolated capacitor)

No battery is connected, so Q is fixed and the agent's work goes entirely into stored energy:

$$F\,\mathrm{d}x = \mathrm{d}U, \qquad U = \frac{Q^2}{2C} = \frac{Q^2 x}{2\varepsilon_0 A}.$$

U is proportional to x at fixed Q, so

$$F = \frac{\mathrm{d}U}{\mathrm{d}x} = \frac{Q^2}{2\varepsilon_0 A}.$$

Dividing by the area and writing $\sigma = Q/A$, $E = \sigma/\varepsilon_0$:

$$p = \frac{F}{A} = \frac{Q^2}{2\varepsilon_0 A^2} = \frac{\sigma^2}{2\varepsilon_0} = \tfrac{1}{2}\varepsilon_0 E^2.$$

Physical meaning. At constant Q the field E between the plates does not change. Separating the plates by $\mathrm{d}x$ creates a new volume $A\,\mathrm{d}x$ of field with energy density $\frac{1}{2}\varepsilon_0 E^2$, and the agent supplies exactly that energy: $F\,\mathrm{d}x = \frac{1}{2}\varepsilon_0 E^2 A\,\mathrm{d}x$.

(b) Constant voltage V (connected to a battery)

Now V is fixed and $Q = CV$ changes as C changes. When the plate moves by $\mathrm{d}x$, the capacitance changes by $\mathrm{d}C$ and the battery pushes charge $\mathrm{d}Q = V\,\mathrm{d}C$ onto the plates, doing work $V\,\mathrm{d}Q = V^2\,\mathrm{d}C$. Energy conservation reads

$$\underbrace{F\,\mathrm{d}x}_{\text{agent's work}} + \underbrace{V^2\,\mathrm{d}C}_{\text{battery's work}} = \mathrm{d}U = \mathrm{d}\left(\tfrac{1}{2}CV^2\right) = \tfrac{1}{2}V^2\,\mathrm{d}C.$$

Rearranging,

$$F\,\mathrm{d}x = \tfrac{1}{2}V^2\,\mathrm{d}C - V^2\,\mathrm{d}C = -\tfrac{1}{2}V^2\,\mathrm{d}C.$$

Since $C = \varepsilon_0 A/x$, $\mathrm{d}C = -\varepsilon_0 A\,\mathrm{d}x/x^2$ (negative: separating the plates reduces C), so

$$F\,\mathrm{d}x = -\tfrac{1}{2}V^2\left(-\frac{\varepsilon_0 A}{x^2}\,\mathrm{d}x\right) \quad\Longrightarrow\quad F = \frac{\varepsilon_0 A V^2}{2x^2} = \tfrac{1}{2}\varepsilon_0\left(\frac{V}{x}\right)^2 A = \tfrac{1}{2}\varepsilon_0 E^2 A,$$

giving again $p = \frac{1}{2}\varepsilon_0 E^2 = \sigma^2/(2\varepsilon_0)$.

The trap. At constant V, $U = \frac{1}{2}CV^2$ *decreases* as x increases. Writing $F = -\,\mathrm{d}U/\,\mathrm{d}x$ would then suggest the plates repel. That is wrong because it leaves out the battery: as the plates separate, charge $|\,\mathrm{d}Q|$ flows back into the battery and the battery *gains* energy $V^2|\,\mathrm{d}C|$, twice the drop $\frac{1}{2}V^2|\,\mathrm{d}C|$ in field energy. In general the force on the plate in the direction of increasing x is

$$F_x = -\left(\frac{\partial U}{\partial x}\right)_Q = +\left(\frac{\partial U}{\partial x}\right)_V,$$

negative here (attraction) in both forms. The same physical force results either way: it depends on the instantaneous charge and geometry, not on what the plates are connected to.

Check from the field acting on the charge

Why $\sigma^2/2\varepsilon_0$ and not $\sigma \times (\sigma/\varepsilon_0)$? The field just outside the upper plate is σ/ε_0, but half of that is produced by the upper plate's own charge (an isolated sheet gives $\sigma/2\varepsilon_0$ on each side), and a charge exerts no net force on itself. The charge on one plate therefore feels only the field of the *other* plate, $\sigma/(2\varepsilon_0)$, so $F = (\sigma A)\cdot\sigma/(2\varepsilon_0)$, as found. Equivalently, the surface charge sits in the *average* of the field just inside the conductor (0) and just outside (σ/ε_0). The same holds on any conductor: there is an outward pressure $p = \sigma^2/(2\varepsilon_0)$ normal to the surface, whatever the sign of σ.

Order of magnitude. At the breakdown field of air, $3\,\mathrm{MV\,m^{-1}}$, $p = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}(8.85\times10^{-12})(3\times10^6)^2 \approx 40\,\mathrm{Pa}$, about 4×10^{-4} atmospheres. Electrostatic forces on macroscopic conductors are weak, which is why electrostatic actuators work best at micrometre scales (MEMS).

Extensions

1. Insert a dielectric slab of permittivity ϵ partly between the plates. Find the force pulling it in, at constant Q and at constant V, and show they agree.
2. Show that the outward pressure on a charged soap bubble is $\sigma^2/(2\varepsilon_0)$ and find the charge needed to change its radius by 1% given surface tension γ.
3. Write the force on a conductor as the integral of the Maxwell stress tensor and show it reproduces $\sigma^2/(2\varepsilon_0)$ normal to the surface.

Problem 12 — Electrostatic balance and its instability

Problem 12 (A) (as set on the sheet)

Two conducting discs of radius r are arranged one above the other with their faces horizontal. The lower disc is fixed and the upper is suspended a distance $a \ll r$ above it by springs. When a potential difference V is applied, the equilibrium separation becomes b. Using the results of question 11, show that the (electrostatic) force between the discs is $V^2\pi r^2\varepsilon_0/(2b^2)$. Show that the system is stable only if $b > 2a/3$.

Supervisor's note

Where the learning is. Stability analysis of an equilibrium: an equilibrium is stable only if the net restoring force grows faster than the destabilising force under a small displacement. The linear spring competes with an attraction that rises as $1/x^2$, so beyond a critical voltage there is no stable equilibrium ("pull-in").

Conceptually tricky. Students find the equilibrium condition easily but are unsure how to test stability. The condition is on the *derivative* of the net force at equilibrium. The result $b > 2a/3$ is independent of the spring constant, voltage and disc size, which surprises most; it is the same physics as pull-in in MEMS switches.

What you need to know before starting

- **Problem 11(b):** the force between plates at fixed potential difference V and separation x is $\varepsilon_0 A V^2/(2x^2)$.
- **Hooke's law:** a spring displaced by δ from its natural length exerts a restoring force $k\delta$. The weight of the disc is already balanced by the springs when $V = 0$, so it does not appear below.
- **Stability of an equilibrium:** displace the system slightly from equilibrium; it is stable if the net force pushes it back. For a net force $G(x)$ near an equilibrium $x = b$, Taylor expansion gives $G(b+\delta) \approx G(b) + G'(b)\,\delta = G'(b)\,\delta$, so only the sign of $G'(b)$ matters.

Final answer

$F = \dfrac{\pi r^2 \varepsilon_0 V^2}{2b^2}$; the equilibrium is stable only for $b > \frac{2}{3}a$, i.e. for $V^2 < V_c^2 = \dfrac{8ka^3}{27\pi\varepsilon_0 r^2}$, where k is the total spring constant.

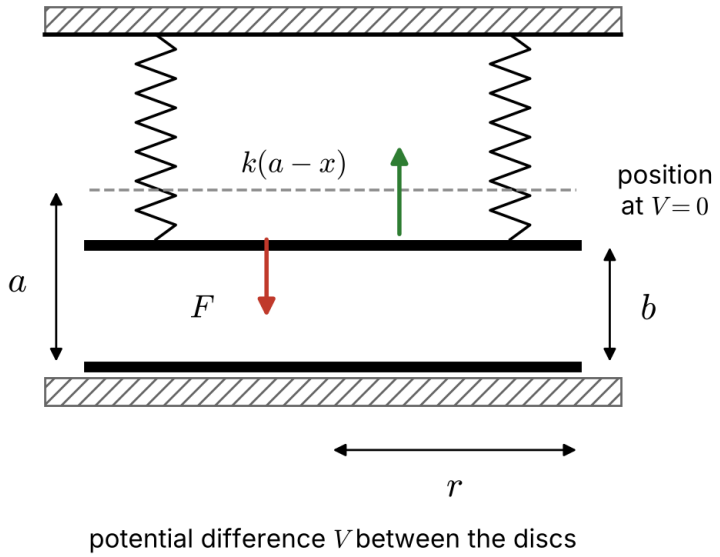


Figure 13: The balance: the upper disc hangs on springs (total constant k), with rest separation a at $V = 0$ and equilibrium separation b at voltage V.

Step 1: the electrostatic force

Because $a \ll r$, the discs form a parallel-plate capacitor of area $A = \pi r^2$ with negligible edge effects. From Problem 11(b), at separation b and voltage V,

$$F = \tfrac{1}{2}\varepsilon_0 E^2 A = \tfrac{1}{2}\varepsilon_0 \left(\frac{V}{b}\right)^2 \pi r^2 = \frac{V^2 \pi r^2 \varepsilon_0}{2b^2}.$$

Step 2: the equilibrium condition

With no voltage the separation is a (the springs are then at their "rest" position for this problem). Let x be the separation. If the upper disc is pulled down to $x < a$, the springs are stretched by $a - x$ and pull it back up with force $k(a - x)$. At equilibrium, $x = b$, the upward spring force balances the downward electrostatic attraction:

$$k(a - b) = \frac{\varepsilon_0 A V^2}{2b^2}. \tag{1}$$

Step 3: stability

Define the net downward force, i.e. the force tending to *close* the gap:

$$G(x) = \underbrace{\frac{\varepsilon_0 A V^2}{2x^2}}_{\text{electric, down}} - \underbrace{k(a - x)}_{\text{spring, up}}.$$

At equilibrium $G(b) = 0$. Now push the disc down slightly, to $x = b + \delta$ with $\delta < 0$. For stability the net force must then be upward, $G < 0$, so that the disc returns. Since $G(b + \delta) \approx G'(b)\,\delta$ and $\delta < 0$, we need $G'(b) > 0$. (Pulling the disc up, $\delta > 0$, gives the same condition.) Differentiating,

$$G'(x) = -\frac{\varepsilon_0 A V^2}{x^3} + k \quad \Longrightarrow \quad \text{stable} \iff k - \frac{\varepsilon_0 A V^2}{b^3} > 0.$$

Eliminate V. From (1), $\varepsilon_0 A V^2 = 2kb^2(a - b)$. Substituting:

$$k - \frac{2kb^2(a-b)}{b^3} > 0 \iff 1 - \frac{2(a-b)}{b} > 0 \iff b > 2a - 2b \iff \boxed{b > \tfrac{2}{3}a}$$

(We divided by $k > 0$ and multiplied by $b > 0$, which do not reverse the inequality.) Physically: the electric force grows as $1/x^2$ when the gap closes, the spring force only linearly; close enough, the electric force always wins.

Graphical picture. Plot the spring force $k(a - x)$ (a straight line, zero at $x = a$) and the electrostatic force $\varepsilon_0 A V^2/(2x^2)$ against x. For small V they cross twice: the crossing at larger x is stable, the other unstable. As V rises the crossings move together and merge where the curve is tangent to the line, at $x = 2a/3$; beyond that there is no equilibrium and the plate snaps shut.

Critical voltage. Setting $b = 2a/3$ in (1), so $a - b = a/3$:

$$V_c^2 = \frac{2kb^2(a-b)}{\varepsilon_0 A} = \frac{2k \cdot \frac{4a^2}{9} \cdot \frac{a}{3}}{\varepsilon_0 \pi r^2} = \frac{8ka^3}{27\pi\varepsilon_0 r^2}.$$

Only one third of the original gap can be closed stably.

Energy view. At constant V the relevant potential is $U_{\text{eff}}(x) = \frac{1}{2}k(a - x)^2 - \frac{1}{2}C(x)V^2$, where the minus sign comes from the battery's work (Problem 11). Differentiating, $\mathrm{d}U_{\text{eff}}/\,\mathrm{d}x = -k(a - x) - \frac{1}{2}V^2\,\mathrm{d}C/\,\mathrm{d}x = -k(a - x) + \varepsilon_0 A V^2/(2x^2) = G(x)$, so equilibria are stationary points of U_{eff}, and stability requires $\mathrm{d}^2U_{\text{eff}}/\,\mathrm{d}x^2 > 0$, which gives the same condition.

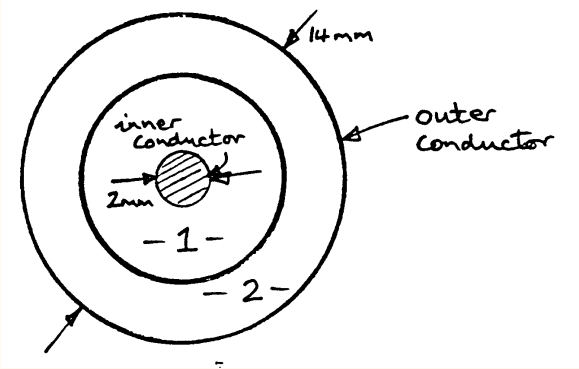
Extensions

1. Repeat at constant *charge*: show the system is then stable for all b. Why does the electrical boundary condition matter so much?
2. Find the frequency of small oscillations about the stable equilibrium as a function of b/a, and show it goes to zero at pull-in.
3. Estimate V_c for a MEMS switch with a 2 µm gap, 100 µm square plate and $k = 1\,\mathrm{N\,m^{-1}}$.

Problem 13 — Coaxial cable with two dielectrics

Problem 13 (A) (as set on the sheet)

A coaxial cable is constructed using two dielectrics as shown in the figure. Dielectric 1 occupies one sixth of the total area between the conductors, whose radii are 1 and 7 mm. The relative permittivities of the dielectrics are $\epsilon_1 = 5$ and $\epsilon_2 = 3$. Calculate the capacitance per unit length of the cable.



Supervisor's note

Where the learning is. Using the electric displacement **D** in layered dielectrics: **D** is fixed by free charge alone, $\mathbf{E} = \mathbf{D}/(\epsilon\varepsilon_0)$ in each layer, and concentric layers act as capacitors in series.

Conceptually tricky. Interpreting "one sixth of the area" as concentric layers (from the figure) and converting area to radius: the interface is at $c = 3$ mm, not at a sixth of the radial distance. Students also sometimes average the permittivities (correct only for dielectrics side by side, in parallel, not for layers in series).

What you need to know before starting

- **Electric displacement.** In a linear dielectric of relative permittivity ϵ, $\mathbf{D} = \epsilon\varepsilon_0\mathbf{E}$, and Gauss's law for **D** involves only *free* charge: $\oint \mathbf{D}\cdot \mathrm{d}\mathbf{A} = Q_{\text{free,enc}}$. The bound (polarisation) charge is taken care of by ϵ.
- **Boundary conditions** at an interface with no free charge: the normal component of **D** and the tangential component of **E** are continuous (derived in Problem 14).
- **Capacitance per unit length** $C' = \lambda/\Delta V$, with $\Delta V = \int \mathbf{E}\cdot \mathrm{d}\mathbf{l}$ from the outer to the inner conductor.
- Note that the sheet's figure gives *diameters* (2 and 14 mm); the radii are 1 and 7 mm.

Final answer

$C' = 111\,\mathrm{pF\,m^{-1}}$ (capacitance per unit length).

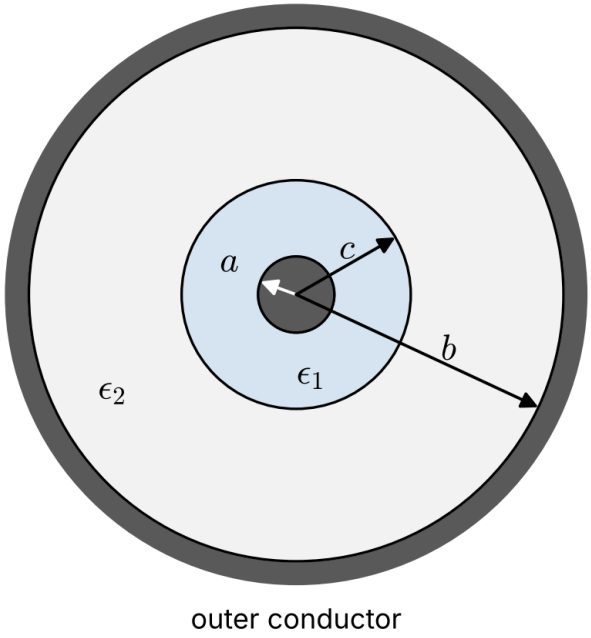


Figure 14: Cross-section of the cable, drawn to scale: inner conductor radius $a = 1$ mm, dielectric 1 out to $c = 3$ mm, dielectric 2 out to $b = 7$ mm.

Set-up. Inner conductor radius $a = 1\,\mathrm{mm}$, outer conductor radius $b = 7\,\mathrm{mm}$. From the figure, dielectric 1 ($\epsilon_1 = 5$) is a concentric layer around the inner conductor, out to a radius c to be found; dielectric 2 ($\epsilon_2 = 3$) fills $c < s < b$, where s is the distance from the axis.

Step 1: locate the interface

The cross-sectional area of an annulus between radii a and c is $\pi(c^2 - a^2)$. Dielectric 1 occupies one sixth of the whole annulus between the conductors:

$$\pi(c^2 - a^2) = \tfrac{1}{6}\pi(b^2 - a^2) \quad \Longrightarrow \quad c^2 = a^2 + \frac{b^2 - a^2}{6} = 1 + \frac{49 - 1}{6} = 9\ \mathrm{mm}^2, \qquad c = 3\,\mathrm{mm}.$$

Step 2: the fields

Let the inner conductor carry free charge λ per unit length (and the outer $-\lambda$). By the cylindrical symmetry, $\mathbf{D}$ is radial and depends only on s (the argument is the same as in Problem 1(d)). Apply Gauss's law for $\mathbf{D}$ to a coaxial cylinder of radius s and length ℓ: only the curved surface has flux, and the free charge enclosed is $\lambda\ell$ whatever the dielectric, so

$$D(s) \cdot 2\pi s\ell = \lambda\ell \quad \Longrightarrow \quad D = \frac{\lambda}{2\pi s} \qquad \text{in both layers.}$$

Then in layer i,

$$E = \frac{D}{\epsilon_i \varepsilon_0} = \frac{\lambda}{2\pi\epsilon_i\varepsilon_0 s}.$$

$\mathbf{D}$ is normal to the interface and continuous across it (no free charge there), consistent with the boundary condition; E jumps by the factor ϵ_1/ϵ_2 at $s = c$.

Step 3: potential difference and capacitance

Integrate E across the gap, splitting at $s = c$, and use $\int \mathrm{d}s/s = \ln s$:

$$\Delta V = \int_a^c \frac{\lambda\,\mathrm{d}s}{2\pi\epsilon_1\varepsilon_0 s} + \int_c^b \frac{\lambda\,\mathrm{d}s}{2\pi\epsilon_2\varepsilon_0 s} = \frac{\lambda}{2\pi\varepsilon_0}\left[\frac{\ln(c/a)}{\epsilon_1} + \frac{\ln(b/c)}{\epsilon_2}\right],$$

$$C' = \frac{\lambda}{\Delta V} = \frac{2\pi\varepsilon_0}{\dfrac{\ln(c/a)}{\epsilon_1} + \dfrac{\ln(b/c)}{\epsilon_2}}.$$

Numerically,

$$\frac{\ln 3}{5} + \frac{\ln(7/3)}{3} = \frac{1.0986}{5} + \frac{0.8473}{3} = 0.2197 + 0.2824 = 0.5022,$$

and, with $2\pi\varepsilon_0 = 5.563 \times 10^{-11}\,\mathrm{F\,m^{-1}}$,

$$C' = \frac{5.563 \times 10^{-11}\,\mathrm{F\,m^{-1}}}{0.5022} = 1.11 \times 10^{-10}\,\mathrm{F\,m^{-1}} = 111\,\mathrm{pF\,m^{-1}}.$$

Series capacitors. The result is $1/C' = 1/C_1' + 1/C_2'$ with $C_1' = 2\pi\epsilon_1\varepsilon_0/\ln(c/a)$ and $C_2' = 2\pi\epsilon_2\varepsilon_0/\ln(b/c)$: the same charge λ sits on both "capacitors" and their voltages add, exactly as for capacitors in series.

Remarks

- **Why put the high-ϵ layer inside?** The field in a single dielectric is largest at the inner conductor ($E \propto 1/s$). A high-permittivity inner layer lowers E there and evens out the field, raising the voltage the cable can carry before breakdown ("grading" of cable insulation).
- **Bound charge.** At the interface **D** is continuous but **E** jumps, so there is a bound surface charge $\sigma_b = \varepsilon_0(E_2 - E_1)$ there, positive for $\lambda > 0$.

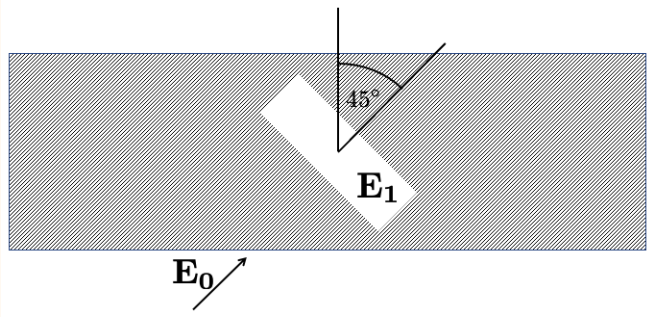
Extensions

1. Find the maximum field in each layer per unit voltage, and choose c to make the two maxima equal.
2. Suppose instead the dielectrics were arranged as sectors (dielectric 1 filling a 60° wedge from a to b). Show the capacitors are then in parallel and find C'.
3. Compute the bound surface charge densities on all three surfaces and check that the total bound charge is zero.

Problem 14 — Field in a tilted cavity in a dielectric slab

Problem 14 (A) (as set on the sheet)

A planar slab of a dielectric material for which $\epsilon = 2$ has a disc-shaped cavity in its interior, as shown from the side in the sketch below. Both slab and cavity are much wider than they are thick, and the normal to the plane of the cavity is at 45° to that of the slab. The slab is placed in a uniform electric field E_0. If the field inside the cavity is E_1, what is the ratio E_1/E_0 when the direction of E_0 is parallel to the normal to the plane of the cavity?



Supervisor's note

Where the learning is. Applying the two boundary conditions (D_n continuous, E_t continuous) in sequence: first at the slab face, then at the cavity face. Thin flat geometries make each interface effectively infinite, so the conditions fix the field completely.

Conceptually tricky. Keeping track of two different normals. Students often apply the cavity boundary condition using the slab normal, or assume the field in the slab is still along $\mathbf{E}_0$. The field direction changes at each interface, and the answer is not simply ϵ or $1/\epsilon$.

What you need to know before starting

- $\mathbf{D} = \epsilon\varepsilon_0\mathbf{E}$ in a linear dielectric ($\epsilon = 1$ in vacuum), and Gauss's law for $\mathbf{D}$ with free charge only (Problem 13).
- **Boundary conditions** at an interface carrying no free charge (derived in Step 0 below): D_n (normal component of $\mathbf{D}$) is continuous; E_t (tangential component of $\mathbf{E}$) is continuous.
- **Why "much wider than thick" matters:** far from the edges of a thin flat region, the field inside is uniform and each face looks like an infinite plane, so the boundary conditions alone fix the field. (For a cavity of general shape one would need to solve Laplace's equation.)
- Resolving a vector into components along two perpendicular unit vectors, $E_n = \mathbf{E}\cdot\hat{\mathbf{n}}$, $E_t = \mathbf{E}\cdot\hat{\mathbf{t}}$.

Final answer

$E_1/E_0 = \sqrt{37}/4 = 1.52$; $\mathbf{E}_1$ lies 9.5° from the cavity normal.

Step 0: where the boundary conditions come from

D_n **continuous.** Take a flat pillbox of face area A straddling the interface, with height $\to 0$. Gauss's law for $\mathbf{D}$: $D_n^{(2)}A - D_n^{(1)}A =$ free charge enclosed $= 0$. So $D_n^{(1)} = D_n^{(2)}$, i.e. $\epsilon^{(1)}E_n^{(1)} = \epsilon^{(2)}E_n^{(2)}$.

E_t **continuous.** Take a thin rectangular loop with long sides of length ℓ on either side of the interface. $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l} = E_t^{(2)}\ell - E_t^{(1)}\ell = 0$ (the short sides contribute nothing as their length $\to 0$). So $E_t^{(1)} = E_t^{(2)}$.

Set-up. Take the slab normal along $\hat{\mathbf{z}}$ and work in the xz-plane (the plane of the sketch). The cavity normal $\hat{\mathbf{n}}_c$ is at 45° to $\hat{\mathbf{z}}$, and $\mathbf{E}_0$ is parallel to it. Using $\cos 45^\circ = \sin 45^\circ = 1/\sqrt{2}$:

$$\hat{\mathbf{n}}_c = \sin 45^\circ\,\hat{\mathbf{x}} + \cos 45^\circ\,\hat{\mathbf{z}} = \tfrac{1}{\sqrt{2}}(1,0,1), \qquad \hat{\mathbf{t}}_c = \tfrac{1}{\sqrt{2}}(1,0,-1), \qquad \mathbf{E}_0 = E_0\,\hat{\mathbf{n}}_c = \frac{E_0}{\sqrt{2}}(1,0,1),$$

where $\hat{\mathbf{t}}_c$ is the unit vector along the cavity face, in the plane of the sketch (check: $\hat{\mathbf{n}}_c \cdot \hat{\mathbf{t}}_c = \frac{1}{2}(1-1) = 0$). The relative permittivity of the slab is $\epsilon = 2$.

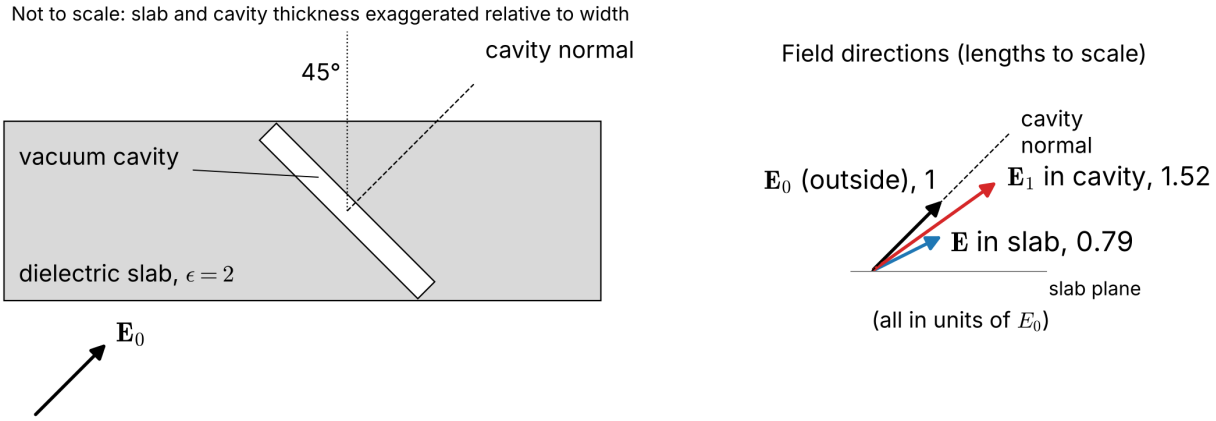


Figure 15: Left: geometry, with thicknesses exaggerated. Right: the three fields drawn to scale from a common origin.

Step 1: field in the slab

At the slab face the normal is $\hat{\mathbf{z}}$, so the normal component is E_z and the tangential one is E_x. Going from vacuum into the dielectric:

$$E_x^{\mathrm{d}} = E_x^0 = \frac{E_0}{\sqrt{2}} \quad (E_t \text{ continuous}), \qquad \epsilon\, E_z^{\mathrm{d}} = 1 \cdot E_z^0 \;\Rightarrow\; E_z^{\mathrm{d}} = \frac{E_0}{\sqrt{2}\,\epsilon} = \frac{E_0}{2\sqrt{2}} \quad (D_n \text{ continuous}).$$

So

$$\mathbf{E}_{\mathrm{d}} = \frac{E_0}{\sqrt{2}}\left(1,\, 0,\, \tfrac{1}{2}\right), \qquad |\mathbf{E}_{\mathrm{d}}| = \frac{E_0}{\sqrt{2}}\sqrt{1+\tfrac{1}{4}} = 0.79\, E_0.$$

The field in the slab is weaker and has been bent away from the slab normal.

Step 2: resolve the slab field along the cavity's directions

The cavity's normal and tangent are different from the slab's, so take dot products with $\hat{\mathbf{n}}_c$ and $\hat{\mathbf{t}}_c$:

$$E_{\mathrm{d},n} = \mathbf{E}_{\mathrm{d}} \cdot \hat{\mathbf{n}}_c = \frac{E_0}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\left(1 \cdot 1 + \tfrac{1}{2} \cdot 1\right) = \frac{E_0}{2} \cdot \frac{3}{2} = \tfrac{3}{4} E_0,$$

$$E_{\mathrm{d},t} = \mathbf{E}_{\mathrm{d}} \cdot \hat{\mathbf{t}}_c = \frac{E_0}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\left(1 \cdot 1 - \tfrac{1}{2} \cdot 1\right) = \frac{E_0}{2} \cdot \frac{1}{2} = \tfrac{1}{4} E_0.$$

Step 3: field in the cavity

At the cavity faces (normal $\hat{\mathbf{n}}_c$), going from dielectric (ϵ) to vacuum (1):

$$1 \cdot E_{1,n} = \epsilon\, E_{\mathrm{d},n} \;\Rightarrow\; E_{1,n} = 2 \times \tfrac{3}{4} E_0 = \tfrac{3}{2} E_0, \qquad E_{1,t} = E_{\mathrm{d},t} = \tfrac{1}{4} E_0.$$

Hence

$$\frac{E_1}{E_0} = \sqrt{\left(\tfrac{3}{2}\right)^2 + \left(\tfrac{1}{4}\right)^2} = \sqrt{\tfrac{36}{16} + \tfrac{1}{16}} = \frac{\sqrt{37}}{4} = 1.52,$$

$$\tan\alpha = \frac{E_{1,t}}{E_{1,n}} = \frac{1/4}{3/2} = \frac{1}{6} \;\Longrightarrow\; \alpha = 9.5^\circ \text{ from } \hat{\mathbf{n}}_c.$$

Remarks

- **Checks by limits.** If the cavity were parallel to the slab (tilt 0°, $\mathbf{E}_0$ along $\hat{\mathbf{z}}$), $D = \varepsilon_0 E_0$ passes straight through both parallel interfaces and $E_1 = E_0$. If the cavity were perpendicular to the slab faces with $\mathbf{E}_0$ along the slab normal, the field in the slab would be E_0/ϵ and tangential to the crack, so $E_1 = E_0/\epsilon$. The tilted case is different from both.
- **Why the field is enhanced.** Thin cavities perpendicular to $\mathbf{D}$ carry the full D into vacuum, so $E = D/\varepsilon_0$ is large there. This is why thin voids in insulation are where partial discharge and breakdown start.

Extensions

1. Find E_1/E_0 as a function of the tilt angle β of the cavity (keeping $\mathbf{E}_0$ along the cavity normal), and plot it from 0 to 90°.
2. Replace the thin disc by a needle-shaped cavity along the same direction, and find E_1. Which shape gives the larger field?
3. Find the bound surface charge densities on the slab faces and on the cavity faces.

Problem 15 — Dielectric relaxation time

Problem 15 (A) (as set on the sheet)

Show that an arbitrary charge distribution in a semiconductor of conductivity σ and permittivity ϵ decays with time constant $\tau = \epsilon\varepsilon_0/\sigma$. [$\tau$ is known as the dielectric relaxation time.] *Hint: don't forget Ohm's Law,* $\mathbf{J} = \sigma\mathbf{E}$, *and charge conservation.*

Supervisor's note

Where the learning is. Combining three laws — charge conservation, Ohm's law and Gauss's law — into a single first-order equation for ρ. Students see that net charge cannot persist inside a uniform conductor and learn the timescale on which a material behaves as a conductor rather than an insulator.

Conceptually tricky. The decay is local: every volume element loses charge at the same rate, independent of the shape of the distribution. There is no spreading or diffusion, and the charge does not "move to" anywhere within the bulk; it appears on the surface. Students also confuse τ with the RC time of a circuit, although for a capacitor filled with the medium they are the same.

What you need to know before starting

- **The continuity equation** $\nabla \cdot \mathbf{J} + \partial\rho/\partial t = 0$ (local charge conservation), derived in Step 1 from the **divergence theorem**, $\oint_S \mathbf{J} \cdot \mathrm{d}\mathbf{A} = \int_V \nabla \cdot \mathbf{J}\, \mathrm{d}\tau$.
- **Ohm's law in local form,** $\mathbf{J} = \sigma\mathbf{E}$ (current density proportional to field).
- **Gauss's law in a dielectric,** $\nabla \cdot \mathbf{D} = \rho_{\text{free}}$ with $\mathbf{D} = \epsilon\varepsilon_0\mathbf{E}$ (Problem 13).
- Solving $\mathrm{d}y/\,\mathrm{d}t = -y/\tau$ by separation of variables.

Final answer

$\rho(\mathbf{r},t) = \rho(\mathbf{r},0)\, e^{-t/\tau}$ with $\tau = \epsilon\varepsilon_0/\sigma$, where ϵ is the relative permittivity and σ the conductivity.

Step 1: charge conservation (the continuity equation)

Take any fixed volume V with closed surface S. The charge inside can only change by current flowing out through S:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\,\mathrm{d}\tau = -\oint_S \mathbf{J}\cdot\,\mathrm{d}\mathbf{A} = -\int_V \nabla\cdot\mathbf{J}\,\mathrm{d}\tau,$$

using the divergence theorem in the last step. This holds for every volume, however small, so the integrands must be equal:

$$\frac{\partial\rho}{\partial t} = -\nabla\cdot\mathbf{J}.$$

Step 2: Ohm's law

With $\mathbf{J} = \sigma\mathbf{E}$ and σ the same everywhere in the material, σ comes out of the divergence:

$$\nabla\cdot\mathbf{J} = \sigma\,\nabla\cdot\mathbf{E}.$$

Step 3: Gauss's law

In a uniform linear dielectric, $\nabla\cdot\mathbf{D} = \rho$ with $\mathbf{D} = \epsilon\varepsilon_0\mathbf{E}$, where ρ is the free charge density:

$$\nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon\varepsilon_0}.$$

Step 4: combine and solve

Substituting Step 3 into Step 2, and the result into Step 1:

$$\frac{\partial\rho}{\partial t} = -\frac{\sigma}{\epsilon\varepsilon_0}\,\rho = -\frac{\rho}{\tau}, \qquad \tau \equiv \frac{\epsilon\varepsilon_0}{\sigma}.$$

At each fixed point $\mathbf{r}$ this is an ordinary differential equation in t. Separate the variables and integrate from 0 to t:

$$\int_{\rho(\mathbf{r},0)}^{\rho(\mathbf{r},t)} \frac{\mathrm{d}\rho}{\rho} = -\int_0^t \frac{\mathrm{d}t'}{\tau} \quad\Longrightarrow\quad \ln\frac{\rho(\mathbf{r},t)}{\rho(\mathbf{r},0)} = -\frac{t}{\tau} \quad\Longrightarrow\quad \rho(\mathbf{r},t) = \rho(\mathbf{r},0)\,e^{-t/\tau}, \qquad \boxed{\tau = \frac{\epsilon\varepsilon_0}{\sigma}}$$

Units check: ε_0 is in $\mathrm{F\,m^{-1}} = \mathrm{s\,\Omega^{-1}\,m^{-1}}$ and σ in $\mathrm{\Omega^{-1}\,m^{-1}}$, so τ is in seconds.

Interpretation

- The equation contains no spatial derivatives, so each point decays independently at the same rate: an arbitrary distribution keeps its shape while its amplitude falls.
- The charge leaving the interior accumulates on the boundary, where σ changes abruptly and the derivation (which assumed uniform σ) no longer applies. The total charge is conserved.
- Equivalently, τ is the RC time of a capacitor filled with the medium: $C = \epsilon\varepsilon_0 A/d$ and $R = d/(\sigma A)$ give $RC = \epsilon\varepsilon_0/\sigma$, independent of geometry.

Orders of magnitude

Material	σ (S m^{-1})	ϵ	τ
Copper	6×10^7	~ 1	1.5×10^{-19} s (see note)
Seawater	5	80	1.4×10^{-10} s
Damp soil	10^{-2}	~ 20	2×10^{-8} s
Intrinsic silicon	4.3×10^{-4}	11.7	2.4×10^{-7} s
Fused quartz	$\sim 10^{-17}$	3.8	$\sim 10^6$ s (weeks)

For a metal the formula predicts a time far shorter than the electron collision time ($\sim 10^{-14}$ s), where Ohm's law in this simple form fails; the true response is a plasma oscillation, damped on the collision time. For damp soil $\tau \sim 20$ ns, which justifies treating the ground as a perfect conductor in Problem 8.

Extensions

1. Include diffusion of charge carriers, $\mathbf{J} = \sigma\mathbf{E} - D_c \nabla\rho$ with D_c the diffusion coefficient. Show that a static charge perturbation is screened over the Debye length $\lambda_D = \sqrt{\epsilon\varepsilon_0 D_c/\sigma}$.
2. A metal sphere is suddenly given a charge at its centre. Estimate, using the corrected picture for metals, how long the charge takes to reach the surface.
3. At what frequency does seawater change from behaving as a conductor to behaving as a dielectric? Relate this to why submarines communicate at very low frequencies.

Magnetostatics

Problem 16 — Long straight wire and the definition of the ampere

Problem 16 (A) (as set on the sheet)

(a) Use the Biot-Savart law to show that the magnetic field at a perpendicular distance r from an infinitely-long, thin wire, carrying a current I, is $B = \mu_0 I/2\pi r$.

(b) Find the magnitude and direction of the force per unit length between two infinitely-long, parallel wires a distance r apart, if one carries a current I_1 and the other I_2. Is this consistent with the interim (1946-2019) definition of the ampere in the SI system of units?

Supervisor's note

Where the learning is. The first application of the Biot–Savart law, the right-hand rule for the direction of $\mathbf{B}$, and the force on a current, $\mathbf{F} = I\mathbf{l} \times \mathbf{B}$. Part (b) links the physics to metrology: how the SI ampere was defined from 1946 to 2019, and why it was redefined.

Conceptually tricky. The geometry of $\mathrm{d}\mathbf{l} \times \mathbf{R}$ (students lose the sin factor or the direction). Many do not realise that the old definition fixed $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} exactly, and that since 2019 μ_0 is a measured quantity because the elementary charge is fixed instead.

What you need to know before starting

- **Biot–Savart law:** a current element $I\,\mathrm{d}\mathbf{l}$ produces $\mathrm{d}\mathbf{B} = \dfrac{\mu_0}{4\pi}\dfrac{I\,\mathrm{d}\mathbf{l} \times \mathbf{R}}{|\mathbf{R}|^3}$ at displacement $\mathbf{R}$

from the element *to* the field point.

- **Cross products of unit vectors:** $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$, $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$, $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$ (cyclic), reversed order gives a minus sign, and $\hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$.
- **Force on a straight current-carrying wire** of length l in a field $\mathbf{B}$: $\mathbf{F} = I\mathbf{l} \times \mathbf{B}$.
- The integral $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}z}{(z^2+r^2)^{3/2}} = \frac{2}{r^2}$, from Problem 1 (with $L \to \infty$).

Final answer

(a) $B = \mu_0 I/(2\pi r)$, circling the wire by the right-hand rule. (b) $F/l = \mu_0 I_1 I_2/(2\pi r)$; attractive for parallel currents, repulsive for antiparallel. Consistent with the 1946–2019 definition, which in fact set $\mu_0/(2\pi) = 2 \times 10^{-7}\,\mathrm{N\,A^{-2}}$ exactly.

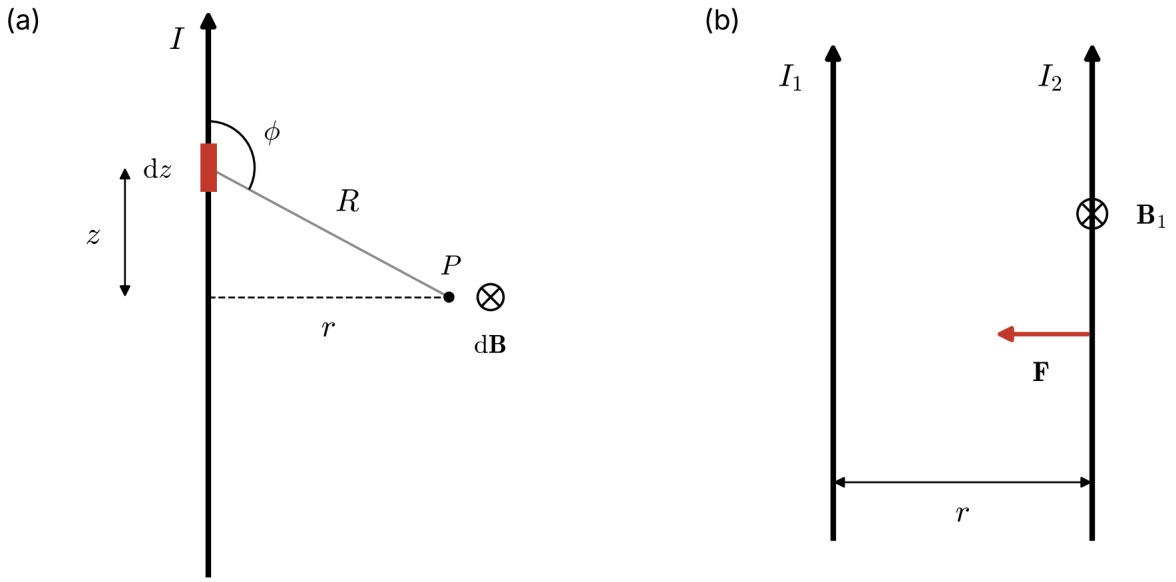


Figure 16: (a) Biot–Savart geometry: the element $\mathrm{d}z$ at height z, the field point P at perpendicular distance r, and ϕ, the angle between $\mathrm{d}\mathbf{l}$ and $\mathbf{R}$; $\mathrm{d}\mathbf{B}$ points into the page. (b) Wire 2 in the field $\mathbf{B}_1$ of wire 1 (into the page) feels a force towards wire 1.

(a) Field of a long straight wire by Biot–Savart

Coordinates. Put the wire along the z-axis carrying current I in the $+z$ direction, and the field point P at $(r, 0, 0)$. An element of the wire at height z is $\mathrm{d}\mathbf{l} = \mathrm{d}z\,\hat{\mathbf{z}}$, and the vector from it to P is

$$\mathbf{R} = (r,0,0) - (0,0,z) = r\,\hat{\mathbf{x}} - z\,\hat{\mathbf{z}}, \qquad |\mathbf{R}| = \sqrt{r^2+z^2}.$$

The cross product. Using $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$ and $\hat{\mathbf{z}} \times \hat{\mathbf{z}} = 0$,

$$\mathrm{d}\mathbf{l} \times \mathbf{R} = \mathrm{d}z\,\hat{\mathbf{z}} \times (r\,\hat{\mathbf{x}} - z\,\hat{\mathbf{z}}) = r\,\mathrm{d}z\,\hat{\mathbf{y}}.$$

So every element gives a field in the *same* direction, $\hat{\mathbf{y}}$, which at P is the azimuthal direction $\hat{\boldsymbol{\phi}}$ round the wire. (Its magnitude is $\mathrm{d}z\,|\mathbf{R}|\sin\phi$ with $\sin\phi = r/|\mathbf{R}|$, the angle in the figure.) The magnitudes simply add:

$$B = \frac{\mu_0 I}{4\pi}\int_{-\infty}^{\infty} \frac{r\,\mathrm{d}z}{(z^2+r^2)^{3/2}} = \frac{\mu_0 I}{4\pi}\cdot r \cdot \frac{2}{r^2} = \frac{\mu_0 I}{2\pi r}.$$

The integral is the one evaluated in Problem 1 by the substitution $z = r\tan\alpha$. The field lines are circles round the wire; their sense is given by the right-hand rule (thumb along I, fingers along $\mathbf{B}$).

Check by Ampère's law. By symmetry $\mathbf{B} = B(r)\hat{\boldsymbol{\phi}}$, so on a circle of radius r, $\oint \mathbf{B}\cdot \mathrm{d}\mathbf{l} = B\cdot 2\pi r = \mu_0 I$, the same result in one line, for the same symmetry reasons as in Problem 1(d).

(b) Force between parallel wires

Let wire 1 lie along the z-axis carrying $I_1\hat{\mathbf{z}}$, and wire 2 be parallel to it at $(r, 0)$ carrying $I_2\hat{\mathbf{z}}$. From (a), wire 1's field at wire 2 is $\mathbf{B}_1 = \dfrac{\mu_0 I_1}{2\pi r}\hat{\mathbf{y}}$. The force on a length l of wire 2 is

$$\mathbf{F} = I_2\, l\,\hat{\mathbf{z}}\times\mathbf{B}_1 = \frac{\mu_0 I_1 I_2 l}{2\pi r}\,(\hat{\mathbf{z}}\times\hat{\mathbf{y}}) = -\frac{\mu_0 I_1 I_2 l}{2\pi r}\,\hat{\mathbf{x}},$$

using $\hat{\mathbf{z}}\times\hat{\mathbf{y}} = -\hat{\mathbf{x}}$. So

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r},$$

directed in $-\hat{\mathbf{x}}$, i.e. *towards* wire 1, when $I_1 I_2 > 0$ (parallel currents attract), and away from it for antiparallel currents. By Newton's third law wire 1 feels an equal and opposite force.

The 1946–2019 ampere. One ampere was the constant current which, in two straight parallel conductors of infinite length and negligible cross-section, 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length. Substituting $I_1 = I_2 = 1$ A, $r = 1$ m:

$$\frac{F}{l} = \frac{\mu_0}{2\pi}\cdot\frac{(1\,\mathrm{A})^2}{1\,\mathrm{m}} = 2\times10^{-7}\,\mathrm{N\,m^{-1}} \quad\Longleftrightarrow\quad \mu_0 = 4\pi\times10^{-7}\,\mathrm{N\,A^{-2}} \text{ exactly.}$$

So the result is consistent, by construction: the definition fixed the value of μ_0. Since May 2019 the ampere is defined by fixing the elementary charge, $e = 1.602\,176\,634\times10^{-19}$ C exactly; μ_0 is now measured and equals $4\pi\times10^{-7}\,\mathrm{N\,A^{-2}}$ to within about one part in 10^{10}.

Extensions

1. Find B at distance r from a *finite* wire segment, and compare with Problem 1(a).
2. Two parallel wires carry 1 kA in opposite directions, 1 cm apart, as in a power busbar. Find the force per metre, and comment on short-circuit forces.
3. Explain why the force between two parallel beams of electrons moving at speed v is reduced by the factor $(1 - v^2/c^2)$ relative to pure electrostatic repulsion.

Problem 17 — Current loop, solenoid, and the force on a dipole

Problem 17 (B) (as set on the sheet)

(a) Show, using the Biot-Savart law, that the magnetic field on the axis of a circular loop carrying a current I, and a distance x away from it, is $B = \mu_0 I a^2/2(a^2+x^2)^{3/2}$, where a is the radius of the loop.

(b) A short magnetic dipole, of moment m, is placed on, and aligned with, the axis of a long solenoid, of radius a, that has n turns per unit length. If the current in the solenoid is I, calculate the force on the dipole when it is outside the solenoid and at a distance x_0 from the end of the solenoid.

Supervisor's note

Where the learning is. (a) Biot–Savart with a symmetry argument that only the axial component survives. (b) Building a solenoid out of loops by integration, and the force on a magnetic dipole in a non-uniform field, $F = m\,\mathrm{d}B/\,\mathrm{d}x$. The result also shows that the end of a long solenoid acts like a magnetic pole.

Conceptually tricky. In (b), setting up the field *outside* a semi-infinite solenoid as an integral over loops (getting the limits right), and recognising that a uniform field exerts no force: the force comes from the field gradient. Students also often expect the field just at the end to equal $\mu_0 nI$; it is half that.

What you need to know before starting

- **Biot–Savart law** (Problem 16): $|\mathrm{d}\mathbf{B}| = \mu_0 I\,|\mathrm{d}\mathbf{l}\times\hat{\mathbf{R}}|/(4\pi R^2)$, perpendicular to both $\mathrm{d}\mathbf{l}$ and $\mathbf{R}$.
- **Additional knowledge for (b):** a small magnetic dipole $\mathbf{m}$ in a field $\mathbf{B}$ has energy $U = -\mathbf{m}\cdot\mathbf{B}$ (the magnetic analogue of $-\mathbf{p}\cdot\mathbf{E}$ in Problem 4), so the force on it is $\mathbf{F} = -\nabla U = \nabla(\mathbf{m}\cdot\mathbf{B})$. A uniform field exerts a torque but no net force. On the axis, the off-axis derivatives of B_x vanish by symmetry, so only $F_x = m\,\mathrm{d}B_x/\,\mathrm{d}x$ survives.
- **A solenoid as a stack of loops:** a length $\mathrm{d}\xi$ of a solenoid with n turns per unit length is a single loop carrying current $nI\,\mathrm{d}\xi$.

Final answer

(a) $B = \dfrac{\mu_0 I a^2}{2(a^2+x^2)^{3/2}}$. (b) $F = \dfrac{\mu_0 n m I a^2}{2(x_0^2+a^2)^{3/2}}$, attracting the dipole towards the solenoid when $\mathbf{m}$ is parallel to the solenoid's field.

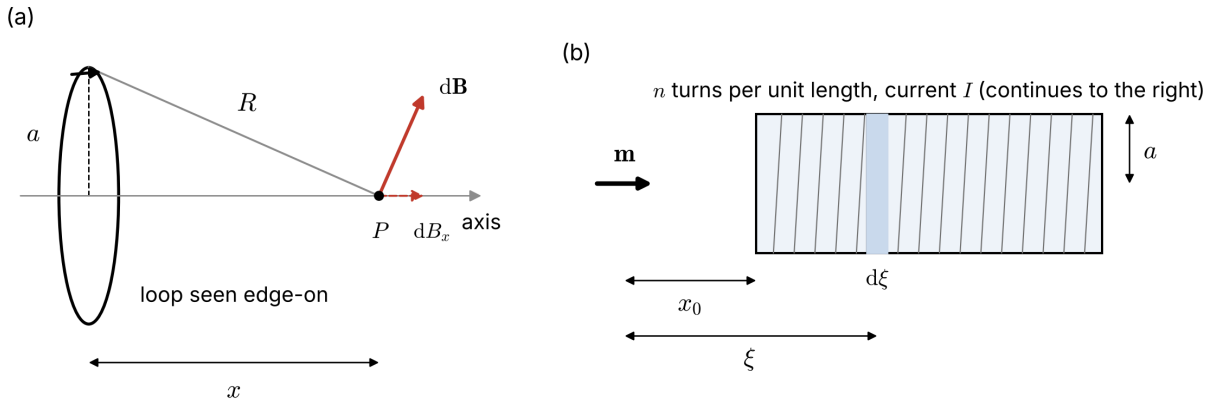


Figure 17: (a) Field on the axis of a loop of radius a, at distance x from its centre: only the axial component $\mathrm{d}B_x$ survives. (b) A dipole $\mathbf{m}$ at distance x_0 from the end of a semi-infinite solenoid, built up from slices of thickness $\mathrm{d}\xi$ at distance ξ.

(a) Field on the axis of a loop

Take the loop of radius a with its axis along x, and a field point P on the axis at distance x from the centre.

Step 1: one element. Every element $\mathrm{d}\mathbf{l}$ of the loop is the same distance $R = \sqrt{a^2+x^2}$ from P, and $\mathrm{d}\mathbf{l}$ (tangent to the loop) is perpendicular to the line $\mathbf{R}$ joining it to P. So $|\mathrm{d}\mathbf{l}\times\hat{\mathbf{R}}| = \mathrm{d}l$, and

by Biot–Savart each element gives a field of size

$$\mathrm{d}B = \frac{\mu_0 I\,\mathrm{d}l}{4\pi R^2},$$

directed perpendicular to both $\mathrm{d}\mathbf{l}$ and $\mathbf{R}$.

Step 2: symmetry. Each $\mathrm{d}\mathbf{B}$ has a part along the axis and a part perpendicular to it. The element on the opposite side of the loop gives the same axial part and an equal and opposite perpendicular part, so the perpendicular parts cancel round the loop and only the axial parts add.

Step 3: the axial part. $\mathrm{d}\mathbf{B}$ is perpendicular to $\mathbf{R}$, so the angle between $\mathrm{d}\mathbf{B}$ and the axis equals the angle between $\mathbf{R}$ and the plane of the loop. Its axial fraction is therefore a/R (radius over hypotenuse in the right-angled triangle formed by the centre, the element and P):

$$\mathrm{d}B_x = \frac{\mu_0 I\,\mathrm{d}l}{4\pi R^2}\cdot\frac{a}{R}.$$

Step 4: add round the loop. R is the same for every element and $\oint \mathrm{d}l = 2\pi a$:

$$B = \frac{\mu_0 I a}{4\pi R^3}\oint \mathrm{d}l = \frac{\mu_0 I a}{4\pi(a^2+x^2)^{3/2}}\cdot 2\pi a = \frac{\mu_0 I a^2}{2(a^2+x^2)^{3/2}}.$$

Checks. At the centre, $B = \mu_0 I/(2a)$. Far away, $x \gg a$, $B \approx \mu_0(I\pi a^2)/(2\pi x^3)$, the axial field of a magnetic dipole of moment $m = I\pi a^2$ (compare the electric $2p/4\pi\varepsilon_0 x^3$).

(b) Force on a dipole near the end of a solenoid

Step 1: the field on the axis outside the end. Let x be the distance of the dipole from the end of a long (effectively semi-infinite) solenoid, measured outwards along the axis. Slice the solenoid into thin rings: the slice at distance ξ from the dipole, of thickness $\mathrm{d}\xi$, is a loop carrying current $nI\,\mathrm{d}\xi$. All the slices produce fields in the same direction along the axis, so by (a), summing from the end ($\xi = x$) to the far end ($\xi \to \infty$),

$$B(x) = \int_x^\infty \frac{\mu_0(nI\,\mathrm{d}\xi)a^2}{2(a^2+\xi^2)^{3/2}} = \frac{\mu_0 nIa^2}{2}\int_x^\infty \frac{\mathrm{d}\xi}{(a^2+\xi^2)^{3/2}}.$$

The integral is the one from Problem 1 with $z \to a$: $\int \mathrm{d}\xi/(a^2+\xi^2)^{3/2} = \xi/(a^2\sqrt{a^2+\xi^2})$. So

$$B(x) = \frac{\mu_0 nI}{2}\left[\frac{\xi}{\sqrt{a^2+\xi^2}}\right]_x^\infty = \frac{\mu_0 nI}{2}\left(1 - \frac{x}{\sqrt{x^2+a^2}}\right),$$

since $\xi/\sqrt{a^2+\xi^2} \to 1$ as $\xi \to \infty$. Check: at $x = 0$, $B = \frac{1}{2}\mu_0 nI$, half the interior field $\mu_0 nI$ of an infinite solenoid, because only half of the infinite solenoid is present (the other half would contribute the same by symmetry).

Step 2: the force. The dipole is aligned with the field, so its energy is $U = -mB(x)$ and the force along the axis (positive meaning away from the solenoid) is $F = -\,\mathrm{d}U/\,\mathrm{d}x = m\,\mathrm{d}B/\,\mathrm{d}x$. Differentiate with the quotient (or product) rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{x}{\sqrt{x^2+a^2}} = \frac{\sqrt{x^2+a^2} - x\cdot\dfrac{x}{\sqrt{x^2+a^2}}}{x^2+a^2} = \frac{(x^2+a^2)-x^2}{(x^2+a^2)^{3/2}} = \frac{a^2}{(x^2+a^2)^{3/2}},$$

so

$$\frac{\mathrm{d}B}{\mathrm{d}x} = -\frac{\mu_0 nIa^2}{2(x^2+a^2)^{3/2}}, \qquad F = m\frac{\mathrm{d}B}{\mathrm{d}x} = -\frac{\mu_0 nmIa^2}{2(x_0^2+a^2)^{3/2}} \quad \text{at } x = x_0.$$

The minus sign means the force points towards decreasing x, into the stronger field at the solenoid: a dipole aligned with the field is attracted, and an antialigned one repelled. Its magnitude is

$$|F| = \frac{\mu_0 nmIa^2}{2(x_0^2+a^2)^{3/2}}.$$

Remarks

- **A neat identity.** The force equals $m \cdot nI$ times the loop-field formula of (a) at distance x_0. Differentiating the integral with respect to its lower limit simply removes the integral, leaving the contribution of the end loop.
- **Magnetic pole picture.** For $x_0 \gg a$, $F \approx \mu_0 nIma^2/(2x_0^3)$. This is exactly the force on the dipole from a magnetic "pole" of strength $q_m = nI\pi a^2$ at the solenoid's end: the end of a long solenoid behaves like a monopole, with the other pole far away.

Extensions

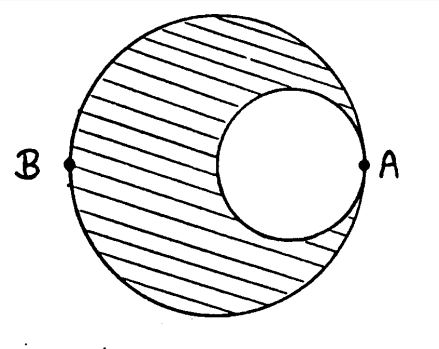
1. Find the field on the axis of a finite solenoid of length ℓ in terms of the angles subtended by its ends, and plot B along the axis.
2. Where on the axis, outside the solenoid, is the force on the dipole largest? (Answer: at the end, $x_0 = 0$.)
3. Show that two coaxial loops separated by their radius (a Helmholtz pair) produce a field with zero first, second and third derivatives at the midpoint.

Problem 18 — Field inside and outside a wire; wire with a cavity

Problem 18 (A) (as set on the sheet)

(a) A long wire of circular cross-section carries a steady current uniformly distributed over its cross-section. Show using Ampère's theorem that the magnetic field varies as r inside the wire, and as $1/r$ outside, where r is the distance from the centre of the wire.

(b) A long wire having the cross-section shown below carries a steady current. The radius of the cavity is half that of the wire. What is the ratio of the magnetic fields at A and B?



Supervisor's note

Where the learning is. Ampère's law for a distributed current, and superposition with "negative current" to handle a hole: the wire with a cavity equals a full wire plus a smaller wire carrying the opposite current density. This is the magnetic analogue of using negative charge for a hole in a charged body.

Conceptually tricky. Students try to apply Ampère's law directly to the wire with the cavity, which fails because the symmetry is broken. Getting the *directions* of the two superposed fields right at A and B is where most errors occur. The surprising bonus result is that the field inside the cavity is uniform.

What you need to know before starting

- **Ampère's law:** $\oint_C \mathbf{B}\cdot\mathrm{d}\mathbf{l} = \mu_0 I_{\text{enc}}$, where I_{enc} is the current through any surface bounded by the loop C. Like Gauss's law, it can be used to *calculate* $\mathbf{B}$ only when symmetry fixes its direction and makes its magnitude constant on the loop.
- **Superposition:** the field of a sum of current distributions is the sum of their fields. A region with no current can be represented as $+\mathbf{J}$ and $-\mathbf{J}$ overlapping.
- **Right-hand rule:** the field of a straight current circles it; with the thumb along the current, the fingers give the sense of $\mathbf{B}$.

Final answer

(a) $B = \mu_0 I r/(2\pi a^2)$ inside, $\mu_0 I/(2\pi r)$ outside. (b) $B_A/B_B = 3/5$.

(a) Solid wire

The wire has radius a and carries current I uniformly, so the current density is $J = I/(\pi a^2)$, along the axis. By symmetry (as in Problem 1(d)), $\mathbf{B} = B(r)\hat{\boldsymbol{\phi}}$: azimuthal, depending only on r. Take an Ampère loop that is a circle of radius r centred on the axis. Then $\mathbf{B}$ is parallel to $\mathrm{d}\mathbf{l}$ and constant in magnitude round it, so $\oint \mathbf{B}\cdot\mathrm{d}\mathbf{l} = B\cdot 2\pi r$. The current through the circle is

$$I_{\text{enc}} = \begin{cases} J\,\pi r^2 = I\,r^2/a^2 & r \le a, \\ I & r \ge a, \end{cases} \qquad \text{so} \qquad B = \begin{cases} \dfrac{\mu_0 I r}{2\pi a^2} & r \le a \quad (\propto r), \\ \dfrac{\mu_0 I}{2\pi r} & r \ge a \quad (\propto 1/r). \end{cases}$$

B rises linearly to a maximum $\mu_0 I/(2\pi a)$ at the surface and then falls as $1/r$.

(b) Wire with a cavity

Geometry (from the sheet's figure). Put the wire's centre O at the origin, with current out of the page ($+\hat{\mathbf{z}}$). The cavity has radius $a/2$ and touches the outer surface at A, so its centre C is at $(a/2,\,0)$. A is at $(a,0)$ and B at $(-a,0)$, both on the outer surface on the line through O and C. Let J be the current density in the metal.

Superposition.

$$\text{wire with cavity} = \text{full wire (radius } a,\ \text{centre O},\ +J) + \text{small wire (radius } a/2,\ \text{centre C},\ -J).$$

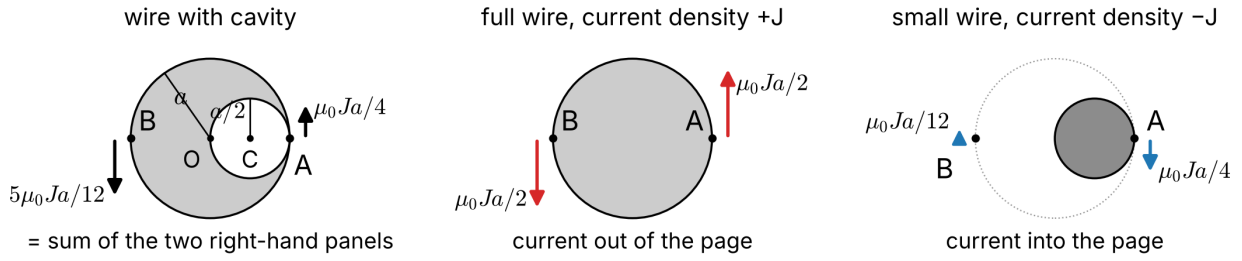


Figure 18: Superposition for Problem 18(b). Arrows show the field directions and magnitudes at A and B; arrow lengths are proportional to field strength.

Step 1: the size of each field. From (a), outside a wire of radius R and current density J, at

distance $d \geq R$ from its axis,

$$B = \frac{\mu_0 (J\pi R^2)}{2\pi d} = \frac{\mu_0 J R^2}{2d}.$$

- **Full wire** ($R = a$): A and B are both at $d = a$, so $B = \frac{1}{2}\mu_0 J a$ at each.
- **Small wire** ($R = a/2$): A is at $d = a/2$, so $B = \mu_0 J(a^2/4)/a = \frac{1}{4}\mu_0 J a$; B is at $d = 3a/2$, so $B = \mu_0 J(a^2/4)/(3a) = \frac{1}{12}\mu_0 J a$.

Step 2: the directions. Each field circles its own wire's axis. A lies to the right of both axes, O and C. By the right-hand rule, at a point to the right of an axis, current out of the page ($+J$) gives a field pointing up the page, and current into the page ($-J$) gives a field pointing down. So at A the two fields are opposite, and the magnitudes subtract. B lies to the left of both axes, so both fields are reversed and again opposite. Hence

$$B_A = \tfrac{1}{2}\mu_0 J a - \tfrac{1}{4}\mu_0 J a = \tfrac{1}{4}\mu_0 J a \ \text{(up)}, \qquad B_B = \tfrac{1}{2}\mu_0 J a - \tfrac{1}{12}\mu_0 J a = \tfrac{5}{12}\mu_0 J a \ \text{(down)},$$

$$\boxed{\frac{B_A}{B_B} = \frac{1/4}{5/12} = \frac{3}{5}}$$

The field is weaker at A, next to the cavity, where the missing current would have contributed most. In terms of the total current, $I = J\pi(a^2 - a^2/4) = \frac{3}{4}\pi a^2 J$, so $B_A = \mu_0 I/(3\pi a)$ and $B_B = 5\mu_0 I/(9\pi a)$.

Remark: the field in the cavity is uniform

Inside a wire, (a) gives a field $\frac{1}{2}\mu_0 J s$ circling the axis, where s is the distance from the axis. As a vector this is $\frac{1}{2}\mu_0 \mathbf{J} \times \mathbf{s}$, with $\mathbf{s}$ the vector from the axis to the point: $\mathbf{J} \times \mathbf{s}$ has magnitude Js and points round the axis in the right-hand sense. At a point inside the cavity, both wires use this inside formula. With $\mathbf{r}$ measured from O, the vector from C is $\mathbf{r} - \mathbf{d}$, where $\mathbf{d} = \frac{a}{2}\hat{\mathbf{x}}$ is the position of C:

$$\mathbf{B}_{\text{cavity}} = \tfrac{1}{2}\mu_0 \mathbf{J} \times \mathbf{r} - \tfrac{1}{2}\mu_0 \mathbf{J} \times (\mathbf{r} - \mathbf{d}) = \tfrac{1}{2}\mu_0 \mathbf{J} \times \mathbf{d},$$

independent of position. With $d = a/2$ this is $\frac{\mu_0 J a}{4}\hat{\mathbf{y}}$ everywhere in the cavity, consistent with the value at A.

Extensions

1. Express the answer in terms of the total current I and find B_A and B_B numerically for $I = 10\,\text{A}$, $a = 1\,\text{mm}$.
2. Find the field at the two points on the outer surface at 90° to the line of centres.
3. Use the uniform-field result to design a region of uniform transverse field (this is the principle of the "cosθ" dipole magnets in particle accelerators).

Problem 19 — Pressure in a current-carrying mercury column (pinch effect)

Problem 19 (B) (as set on the sheet)

A cylindrical column of mercury of radius a carries a current I, uniformly distributed over its cross-section. By considering the force per unit volume on the current, derive a formula for the pressure p as a function of radius r in the column ignoring the weight of the mercury. If $a = 5\,\text{mm}$ and $I = 100\,\text{A}$, what is the difference between the pressure at the centre and that on the edge of the column?

Supervisor's note

Where the learning is. The magnetic force density on a current, $\mathbf{f} = \mathbf{J} \times \mathbf{B}$, and the balance $\nabla p = \mathbf{J} \times \mathbf{B}$ for a conducting fluid (magnetohydrostatics). It introduces the pinch effect, the basis of Z-pinch plasma confinement and of instabilities in liquid-metal systems.

Conceptually tricky. Recognising that the force is radially *inward* for any current direction (the self-field always pinches), and that the boundary condition sets the pressure at the edge, not at the centre. Students also confuse the force per unit volume with the force per unit length.

What you need to know before starting

- **Force density on a current:** a volume $\mathrm{d}\tau$ carrying current density $\mathbf{J}$ is a current element $\mathbf{J}\,\mathrm{d}\tau$, so it feels force $\mathbf{J} \times \mathbf{B}\,\mathrm{d}\tau$; the force per unit volume is $\mathbf{f} = \mathbf{J} \times \mathbf{B}$.
- **The field inside the column** is $B = \mu_0 I r/(2\pi a^2)$, azimuthal (Problem 18(a)).
- **Additional knowledge (fluid statics):** a fluid at rest with pressure $p(\mathbf{r})$ exerts a net force $-\nabla p$ per unit volume on each element (pressure pushes from high to low). Equilibrium with a body force $\mathbf{f}$ requires $-\nabla p + \mathbf{f} = 0$, i.e. $\nabla p = \mathbf{f}$. For a radial dependence, $\nabla p = (\mathrm{d}p/\mathrm{d}r)\,\hat{\mathbf{r}}$.

Final answer

$p(r) = p_0 + \dfrac{\mu_0 I^2 (a^2 - r^2)}{4\pi^2 a^4}$, where p_0 is the pressure at the edge; for $a = 5\,\mathrm{mm}$, $I = 100\,\mathrm{A}$, $\Delta p = 12.7\,\mathrm{Pa}$.

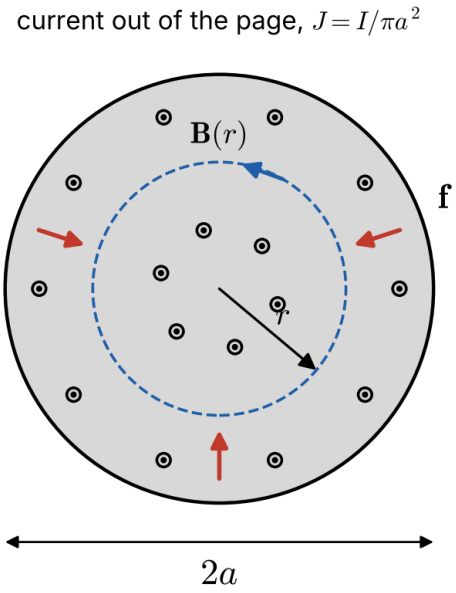


Figure 19: Cross-section of the mercury column (radius a, current out of the page). The self-field $\mathbf{B}(r)$ circulates, and the force density $\mathbf{f} = \mathbf{J} \times \mathbf{B}$ points inwards.

Step 1: force density

Use cylindrical coordinates with the current along $+\hat{\mathbf{z}}$: $\mathbf{J} = J\hat{\mathbf{z}}$ with $J = I/(\pi a^2)$, and $\mathbf{B} = B(r)\hat{\boldsymbol{\phi}}$ with $B(r) = \mu_0 I r/(2\pi a^2)$. The cross product of the unit vectors, using their Cartesian forms

(toolkit),

$$\hat{\mathbf{z}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \times (-\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}) = -\sin\phi\,\hat{\mathbf{y}} - \cos\phi\,\hat{\mathbf{x}} = -\hat{\mathbf{r}},$$

where $\hat{\mathbf{r}} = \cos\phi\,\hat{\mathbf{x}} + \sin\phi\,\hat{\mathbf{y}}$ is the outward radial unit vector. So

$$\mathbf{f} = \mathbf{J} \times \mathbf{B} = -JB(r)\,\hat{\mathbf{r}} = -\frac{I}{\pi a^2} \cdot \frac{\mu_0 I r}{2\pi a^2}\,\hat{\mathbf{r}} = -\frac{\mu_0 I^2 r}{2\pi^2 a^4}\,\hat{\mathbf{r}},$$

radially *inward*. Reversing the current reverses both $\mathbf{J}$ and $\mathbf{B}$, so $\mathbf{f}$ is unchanged: the column always pinches.

Step 2: pressure balance

In equilibrium, ignoring gravity, $\nabla p = \mathbf{f}$. Only the radial component is non-zero:

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{\mu_0 I^2 r}{2\pi^2 a^4}.$$

Integrate from the edge, where the pressure is $p(a) = p_0$ (set by whatever surrounds the column), inwards to radius r:

$$p(r) - p_0 = \int_a^r \left(-\frac{\mu_0 I^2 r'}{2\pi^2 a^4}\right) \mathrm{d}r' = -\frac{\mu_0 I^2}{2\pi^2 a^4} \cdot \frac{r^2 - a^2}{2} \quad \Longrightarrow \quad p(r) = p_0 + \frac{\mu_0 I^2}{4\pi^2 a^4}\,(a^2 - r^2).$$

The pressure is highest on the axis. The difference between centre and edge is

$$\Delta p = p(0) - p_0 = \frac{\mu_0 I^2}{4\pi^2 a^2} = \frac{(4\pi \times 10^{-7})(100)^2}{4\pi^2 (5 \times 10^{-3})^2}\ \mathrm{Pa} = \frac{1.257 \times 10^{-2}}{9.87 \times 10^{-4}}\ \mathrm{Pa} = 12.7\,\mathrm{Pa}.$$

Remarks

- **Magnetic pressure.** At the edge $B_a = \mu_0 I/(2\pi a) = 4\,\mathrm{mT}$, and $B_a^2/(2\mu_0) = 6.4\,\mathrm{Pa}$. So $\Delta p = 2 \times B_a^2/(2\mu_0)$, and the *average* excess pressure over the cross-section equals $B_a^2/(2\mu_0)$: the self-field squeezes the column like a gas of pressure $B^2/(2\mu_0)$ acting from outside.
- **Size.** 12.7 Pa equals the hydrostatic pressure of about 0.1 mm of mercury, so gravity would dominate in this example. But $\Delta p \propto I^2/a^2$: at 100 kA in the same column it would be 127 bar.
- **Instability.** If the column narrows locally, B_a and hence the pinch pressure rise there, narrowing it further: the "sausage" instability. It limits Z-pinch fusion devices and affects liquid-metal batteries and aluminium smelting cells.

Extensions

1. For a plasma of temperature T and number density n, show that pressure balance gives the Bennett relation $\mu_0 I^2 = 8\pi N k_B T$, where N is the number of particles per unit length.
2. Find the current at which the pinch pressure at the centre of the mercury column equals atmospheric pressure.
3. Repeat for a hollow cylindrical shell of current: where does the pressure change occur?

Problem 20 — Paramagnetic sphere in a uniform field

Problem 20 (B) (as set on the sheet)

A paramagnetic sphere of radius a and relative permeability μ is placed in a vacuum in a uniform magnetic field B. Show that the total magnetic moment m induced in the sphere is given by $\mu_0 m = 4\pi a^3(\mu-1)B/(\mu+2)$. If two such spheres are a distance d apart ($d \gg a$) along the direction of B, what is the force between them?

Supervisor's note

Where the learning is. Solving a magnetostatic boundary-value problem with a magnetic scalar potential, using the same mathematics as the conducting sphere (Problem 7) and the dielectric sphere. Students meet the demagnetising factor ($\frac{1}{3}$ for a sphere) and see why the internal $\mathbf{H}$ is reduced. The second part reuses dipole–dipole interactions from Problem 4.

Conceptually tricky. Choosing the right boundary conditions: tangential $\mathbf{H}$ and normal $\mathbf{B}$ are continuous. Students often assume $\mathbf{H}$ inside equals the applied $\mathbf{H}_0$, i.e. ignore demagnetisation. In the force calculation the sign matters: aligned along the field, the moments are head-to-tail and attract, even for diamagnetic spheres.

What you need to know before starting

- **Additional knowledge (magnetic materials):** $\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M})$, where $\mathbf{M}$ is the magnetisation (magnetic moment per unit volume). In a linear medium $\mathbf{B} = \mu\mu_0\mathbf{H}$, so $\mathbf{M} = (\mu-1)\mathbf{H}$. A uniformly magnetised sphere has total moment $m = \frac{4}{3}\pi a^3 M$.
- **Magnetic scalar potential.** With no free currents, $\nabla\times\mathbf{H} = 0$, so $\mathbf{H} = -\nabla\phi_m$. Together with $\nabla\cdot\mathbf{B} = 0$ and $\mathbf{B} = \mu\mu_0\mathbf{H}$ (μ uniform) this gives Laplace's equation $\nabla^2\phi_m = 0$, inside and outside.
- **Boundary conditions** at the surface (same derivation as Problem 14, with $\mathbf{B}$ in place of $\mathbf{D}$ and $\mathbf{H}$ in place of $\mathbf{E}$): normal $\mathbf{B}$ continuous; tangential $\mathbf{H}$ continuous (no surface free current).
- **The magnetic dipole potential** $\phi_m = m\cos\theta/(4\pi r^2)$, the analogue of $p\cos\theta/(4\pi\varepsilon_0 r^2)$; and the dipole–dipole energy $U = \dfrac{\mu_0}{4\pi d^3}\left[\mathbf{m}_1\cdot\mathbf{m}_2 - 3(\mathbf{m}_1\cdot\hat{\mathbf{d}})(\mathbf{m}_2\cdot\hat{\mathbf{d}})\right]$, the analogue of Problem 4 with $1/\varepsilon_0 \to \mu_0$.

Final answer

$\mu_0\mathbf{m} = \dfrac{4\pi a^3(\mu-1)}{\mu+2}\,\mathbf{B}$; the spheres attract with force $F = \dfrac{24\pi a^6 B^2(\mu-1)^2}{\mu_0(\mu+2)^2 d^4}$.

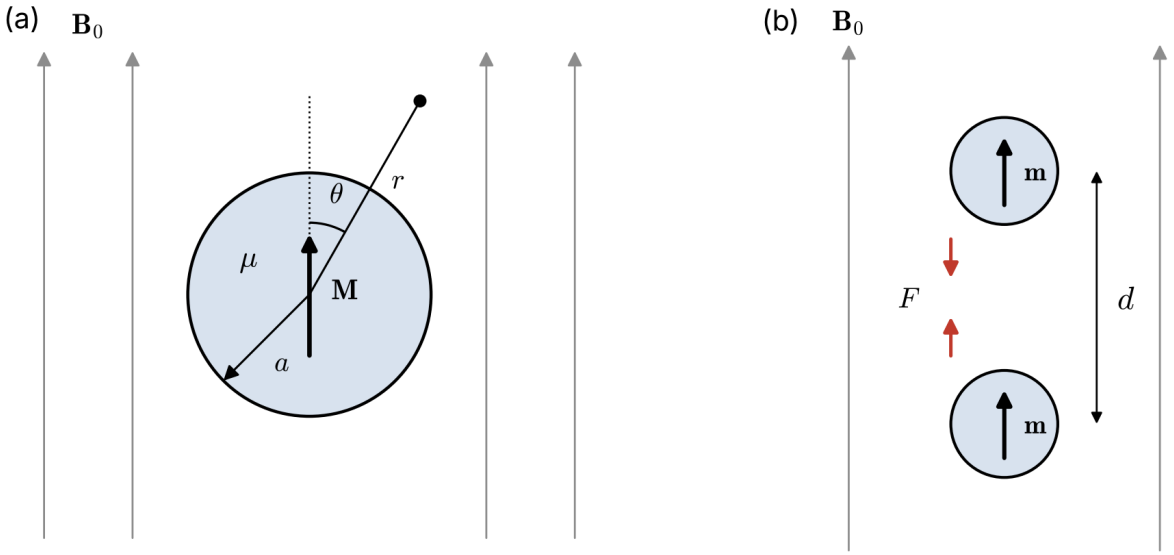


Figure 20: (a) A magnetisable sphere of radius a in the uniform field $\mathbf{B}_0$, with the polar coordinates r, θ used outside. (b) Two such spheres aligned along $\mathbf{B}_0$, a distance d apart; the force F is attractive.

Set-up. The applied field is $\mathbf{B} = \mu_0 \mathbf{H}_0$ along z, with $H_0 = B/\mu_0$. Use spherical polars centred on the sphere with θ measured from $+z$. A uniform field $H_0\hat{\mathbf{z}}$ has potential $\phi_m = -H_0 z = -H_0 r\cos\theta$ (check: $-\partial\phi_m/\partial z = H_0$).

Step 1: trial solution

As for the conducting sphere in Problem 7, try

$$\phi_{\text{out}} = -H_0 r\cos\theta + \frac{A\cos\theta}{r^2}, \qquad \phi_{\text{in}} = -C\,r\cos\theta,$$

with constants A and C to be found. Outside: the applied field plus a dipole (which dies away at large r). Inside: a uniform field, since $-Cr\cos\theta = -Cz$ gives $\mathbf{H}_{\text{in}} = C\hat{\mathbf{z}}$. (A $1/r^2$ term is not allowed inside because it would blow up at the centre.) All three terms satisfy Laplace's equation.

Step 2: boundary conditions at $r = a$

Tangential H continuous. $H_\theta = -\frac{1}{r}\partial\phi_m/\partial\theta$; if this matches for all θ on $r = a$, then ϕ_m itself is continuous there (up to a constant, which we set to zero). Cancelling the common factor $\cos\theta$:

$$-H_0 a + \frac{A}{a^2} = -Ca. \tag{2}$$

Normal B continuous. $B_r = \mu_{\text{rel}}\mu_0 H_r$ with $H_r = -\partial\phi_m/\partial r$:

$$H_r^{\text{out}} = H_0\cos\theta + \frac{2A\cos\theta}{r^3}, \qquad H_r^{\text{in}} = C\cos\theta.$$

At $r = a$, $\mu_0 H_r^{\text{out}} = \mu\mu_0 H_r^{\text{in}}$, i.e. (cancelling $\mu_0\cos\theta$)

$$H_0 + \frac{2A}{a^3} = \mu C. \tag{3}$$

Solve. From (2), dividing by $-a$: $C = H_0 - A/a^3$. Substitute in (3):

$$H_0 + \frac{2A}{a^3} = \mu H_0 - \frac{\mu A}{a^3} \quad\Longrightarrow\quad \frac{A}{a^3}(2+\mu) = (\mu-1)H_0 \quad\Longrightarrow\quad A = \frac{\mu-1}{\mu+2}\,a^3 H_0,$$

$$C = H_0 - \frac{\mu-1}{\mu+2}H_0 = \frac{(\mu+2)-(\mu-1)}{\mu+2}H_0 = \frac{3H_0}{\mu+2}.$$

Step 3: the induced moment

Outside, the extra term $A\cos\theta/r^2$ is exactly the potential $m\cos\theta/(4\pi r^2)$ of a point magnetic dipole at the centre, so $m = 4\pi A$:

$$m = \frac{4\pi a^3(\mu-1)}{\mu+2}\,H_0 \quad\Longrightarrow\quad \boxed{\mu_0\mathbf{m} = \frac{4\pi a^3(\mu-1)}{\mu+2}\,\mathbf{B}}$$

using $\mu_0 H_0 = B$.

Check via magnetisation. Inside, $H_{\text{in}} = C = 3H_0/(\mu+2)$, uniform and *smaller* than H_0 for $\mu > 1$. The magnetisation is $M = (\mu-1)H_{\text{in}} = 3(\mu-1)H_0/(\mu+2)$, and $m = \frac{4}{3}\pi a^3 M = 4\pi a^3(\mu-1)H_0/(\mu+2)$, the same result. One can also check that $H_{\text{in}} = H_0 - \frac{1}{3}M$: the sphere's own magnetisation produces an internal field $-\frac{1}{3}\mathbf{M}$ opposing it (the "demagnetising factor" $\frac{1}{3}$).

Step 4: force between two spheres along B

For $d \gg a$, treat each sphere as a point dipole $\mathbf{m} = m\hat{\mathbf{z}}$, and the line joining them is also along $\hat{\mathbf{z}}$, so $\hat{\mathbf{d}} = \hat{\mathbf{z}}$. Then $\mathbf{m}_1\cdot\mathbf{m}_2 = m^2$ and $(\mathbf{m}_1\cdot\hat{\mathbf{d}})(\mathbf{m}_2\cdot\hat{\mathbf{d}}) = m^2$, and the interaction energy is

$$U(d) = \frac{\mu_0}{4\pi d^3}\left[m^2 - 3m^2\right] = -\frac{\mu_0 m^2}{2\pi d^3}$$

(the magnetic version of the head-to-tail case in Problem 4). The force along the line, positive meaning repulsion (increasing d), is

$$F = -\frac{\mathrm{d}U}{\mathrm{d}d} = -\frac{\mu_0 m^2}{2\pi}\cdot\frac{3}{d^4} = -\frac{3\mu_0 m^2}{2\pi d^4}.$$

The minus sign means **attraction**. Substituting $m = 4\pi a^3(\mu-1)B/[\mu_0(\mu+2)]$, so $m^2 = 16\pi^2 a^6(\mu-1)^2B^2/[\mu_0^2(\mu+2)^2]$:

$$|F| = \frac{3\mu_0}{2\pi d^4}\cdot\frac{16\pi^2 a^6(\mu-1)^2B^2}{\mu_0^2(\mu+2)^2} = \frac{24\pi a^6 B^2(\mu-1)^2}{\mu_0(\mu+2)^2 d^4}.$$

Each sphere's field also slightly changes the other's moment; this mutual induction changes the force by a relative amount of order a^3/d^3, negligible for $d \gg a$.

Remarks

- **Limits.** For $\mu \to 1$, $m \approx (\mu-1)\cdot\frac{4}{3}\pi a^3 H_0$: weak magnetisation, no demagnetising correction. For $\mu \to \infty$ (soft iron), $\mu_0 m \to 4\pi a^3 B$, the same form as the conducting sphere's $p = 4\pi\varepsilon_0 a^3 E$: the moment saturates at a value set by geometry alone.
- **Diamagnets.** For $\mu < 1$ both moments point against $\mathbf{B}$, but they are still parallel to each other and head-to-tail, so they still attract along the field. Side by side (perpendicular to $\mathbf{B}$) the spheres repel. This is why particles in magnetorheological and electrorheological fluids form chains along the field.
- **Size of the effect.** For a typical paramagnet $\mu - 1 \sim 10^{-5}$, so the force is $\sim 10^{-10}$ of that between iron spheres of the same size.

Extensions

1. Find the force when the line joining the spheres is perpendicular to $\mathbf{B}$, and the angle at which it vanishes ($\cos^2\theta = 1/3$, $\theta \approx 54.7^\circ$).
2. Find the field everywhere outside the sphere, and the bound surface current density $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ that produces it.
3. Compare with a dielectric sphere in a uniform electric field: show that $(\mu - 1)/(\mu + 2)$ becomes $(\epsilon - 1)/(\epsilon + 2)$, and relate this to the Clausius–Mossotti relation.

Further remarks on the whole set

Summary of answers and methods

	Result	Main method
1	$E = 2\lambda L/[4\pi\varepsilon_0 z\sqrt{z^2+L^2}]$; $\to Q/(4\pi\varepsilon_0 z^2)$; $\to \lambda/(2\pi\varepsilon_0 z)$	Direct integration; Gauss's law
2	$5.56\times10^{-8}\,\mathrm{C}$, $50\,\mathrm{kV\,m^{-1}}$; $2.5\,\mathrm{kV}$, $12.5\,\mathrm{kV\,m^{-1}}$; Earth $\approx -4.6\times10^{5}\,\mathrm{C}$	Isolated-sphere capacitance
3	$\mathbf{E} = p(3xz, 3yz, 3z^2-r^2)/(4\pi\varepsilon_0 r^5)$	Multipole expansion; $\mathbf{E}=-\nabla V$
4	$U = -p^2[3\cos(\theta_1+\theta_2)+\cos(\theta_1-\theta_2)]/(8\pi\varepsilon_0 d^3)$	Axial and equatorial dipole fields
5	$U = 3Q^2/(20\pi\varepsilon_0 a)$; $a = 1.69\,\mathrm{fm}$	Energy integral, three ways
6	Proofs	Gauss's law plus $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{l}=0$
7	Force $\times(1+16a^3/d^3)$	Induced dipole $p=4\pi\varepsilon_0 a^3 E_0$
8	$12.8\,\mathrm{kV\,m^{-1}}$ upwards; zero at $11.0\,\mathrm{km}$	Images in a plane
9	$(5-4/\sqrt{3})\,q^2/(16\pi\varepsilon_0 x^2)$ towards the edge	Five images in a wedge
10	$2.8\,\mu\mathrm{J}$, independent of R	Charge and energy conservation; RC circuit
11	$p = \sigma^2/(2\varepsilon_0) = \frac{1}{2}\varepsilon_0 E^2$	Virtual work, with battery work at constant V
12	$F = V^2\pi r^2\varepsilon_0/(2b^2)$; stable for $b > 2a/3$	Stability of an equilibrium
13	$111\,\mathrm{pF\,m^{-1}}$	$\mathbf{D}$ in layered dielectrics; series capacitors
14	$E_1/E_0 = \sqrt{37}/4 = 1.52$	Boundary conditions at two interfaces
15	$\tau = \epsilon\varepsilon_0/\sigma$	Continuity + Ohm + Gauss
16	$B = \mu_0 I/(2\pi r)$; $F/l = \mu_0 I_1 I_2/(2\pi r)$	Biot–Savart; force on a current
17	$B = \mu_0 I a^2/[2(a^2+x^2)^{3/2}]$; $F = \mu_0 nmIa^2/[2(x_0^2+a^2)^{3/2}]$	Loops summed into a solenoid; $F = m\,\mathrm{d}B/\mathrm{d}x$
18	$B \propto r$ inside, $1/r$ outside; $B_A/B_B = 3/5$	Ampère's law; superposition of $\pm J$
19	$\Delta p = \mu_0 I^2/(4\pi^2 a^2) = 12.7\,\mathrm{Pa}$	$\nabla p = \mathbf{J}\times\mathbf{B}$
20	$\mu_0 m = 4\pi a^3(\mu-1)B/(\mu+2)$; $F \propto d^{-4}$, attractive	Scalar potential; dipole–dipole force

Notes on the printed answers

All printed answers for problems 1–20 are confirmed by independent calculation. Two need qualification:

- **2(a):** the printed $5.5606 \times 10^{-8}\,\mathrm{C}$ uses $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$; CODATA gives $5.5633 \times 10^{-8}\,\mathrm{C}$. Quote $5.56 \times 10^{-8}\,\mathrm{C}$.
- **2(b):** the printed value is a magnitude; the downward fair-weather field means the Earth's charge is negative.

Common pitfalls

- Using Gauss's law to *calculate* $\mathbf{E}$ without the symmetry to support it (1(d) works; 1(a) does not).
- Claiming Gauss's law alone proves 6(b) or 6(c): the tangential condition and the field-free cavity need $\oint \mathbf{E} \cdot \mathrm{d}\mathbf{l} = 0$ or uniqueness.
- Sign errors with induced or image charges: a charge and its induced charge on a conductor always attract.
- Dropping the $\frac{1}{2}$ in $U = \frac{1}{2}\int \rho V \,\mathrm{d}\tau$ (double counting), or wrongly halving image *forces*.
- Subtracting nearly equal energies numerically instead of deriving $\frac{1}{2}C_s(\Delta V)^2$ first.
- Taking $F = -\,\mathrm{d}U/\,\mathrm{d}x$ at constant voltage without the battery's work (11, 12), which reverses the sign of the force.
- Applying a boundary condition with the wrong normal when there are two interfaces (14).
- Assuming $\mathbf{H}$ inside a magnetised body equals the applied field, i.e. ignoring demagnetisation (20).